

COURSE REVIEW

Nutrition Review

Exam-led theory, original book diagrams and corrected past-paper explanations

Updated 15 September 2026 - past-paper teaching cross-reference

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SCOPE

A detailed synthesis of the available general-nutrition exam topics and selected textbook mechanisms, with a separate oral-health supplement. Includes source-by-source question accounting and clearly marked uncertainty. This is not an official syllabus or a claim that all 866 textbook pages are assigned.

How to read this book

	Key idea	The idea to hold on to. Consolidates the section.
	Exam trap	A common wrong answer or transcription error, and the correction.
	Check the source	Evidence is missing or ambiguous. Check the original before relying on it.
	Clinical link	Why it matters at the bedside; often a book extension.
1	Check yourself	Short retrieval questions with the answer directly below. Cover the answer, then test yourself.

Lecture-grounded material and book extensions are distinguished in each section's source line. This is a detailed review of the available sources, not an official exam blueprint.

Original labelled figures remain important. Numerical laboratory ranges and procedures depend on the course method and local SOP.

Original lecture transcripts are embedded as PDF attachments where available. Automated transcripts can contain errors; study the checked synthesis first. Personal study only: original book and lecture illustrations retain their owners' rights.

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01 / Exam-led reading map and evidence

Source basis Local source inventory and editorial study order

This volume adds nutrition to your Term 2 review library. It follows the general-nutrition past-paper material in Downloads, then explains the underlying mechanisms from your local textbook. It is not a transcript of lectures that were not supplied, and a past question does not establish the current examination blueprint.

KEY IDEA

Past-paper teaching update - 15 September 2026

All 128 source items have been cross-referenced again. Eight topic groups now have expanded teaching in their existing chapters, including two additional original book diagrams. See section 30 for the change checklist. The 36 oral-health items remain a separately marked supplement, and unresolved source keys remain unresolved.

SOURCE CODE	WHAT WAS FOUND	HOW IT IS USED
F	nutrition final exam.pdf - 24 screenshot pages	English TUMS exam interface; one visible question per page. Student selections are not treated as correct answers.
N	Nutrition .pdf - 28 scan pages	Mixed English/Persian general-nutrition exam screenshots. Two obvious repeated stems are retained in the map but not counted as new concepts.
D	Taghziyeh.docx (Persian filename) - 40 numbered bilingual questions	General nutrition practice/past-paper compilation; origin and official answer authority unconfirmed.
O	Taghziyeh.pdf (Persian filename) - 36 questions on 10 pages	A separate paper explicitly titled Nutrition in Oral Health. Supplementary, with course relevance unconfirmed.
B	Nutrition & Diet Therapy, 9th ed. - 866 PDF pages	Local 2016 book. Original diagrams and precise PDF page references support this synthesis; current guidance corrects consequential dated statements.

PRIORITY	READ IN THE LOCAL BOOK	WHY
1 - repeated core	Chs 1-4: overview/DRI, carbohydrate, lipid, protein; Chs 6, 8, 9: energy, vitamins, minerals	Classification, calculations, deficiency syndromes and life-stage recommendations dominate the supplied general questions.
2 - life stages	Chs 11-13: pregnancy/lactation, infancy/childhood, ageing	Pregnancy iron/folate/energy/weight gain; feeding safety; B12, D, hydration and lean mass in older adults.
3 - application	Ch 14 assessment; Chs 21-23 diabetes, CVD and renal disease	These topics occur particularly in N. Read principles, not unsupervised treatment protocols.

PRIORITY	READ IN THE LOCAL BOOK	WHY
4 - supplementary	Ch 5 digestion; Ch 17 Nutrition in Practice: oral health (PDF 512-516)	Digestion explains absorption questions. Oral health has its own clearly separated supplement.

CHECK THE SOURCE**What “covered” means**

Every visible source item is accounted for in the three general-paper maps and the oral-health map. Some receive a verified explanation; some are explicitly unresolved or corrected. Accounting for an item is not the same as certifying its marked option. No official syllabus, nutrition lecture slides, or lecture transcript was located by name in the inspected folders.

Study the teaching sections first. Then cover the explanation column in the source maps and answer aloud or in writing. Finally use the integrated recall cases; these are newly written practice, not past-exam questions. Source codes throughout refer to PDF page counters or the numbered DOCX question, not the sometimes inconsistent screen question number.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 7-12, 23-32 (PDF counter, not printed folio) · F - nutrition final exam.pdf - 24 English screenshot pages - All 24 pages; question evidence, not an official answer key · N - Nutrition .pdf - 28 general-nutrition scan pages - All 28 pages; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Questions 1-40; question evidence, not an official answer key · O - Taghziyeh.pdf - Nutrition in Oral Health paper - Questions 1-36; question evidence, not an official answer key

02 / Nutrients, DRIs and food choices

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

The six nutrient classes are carbohydrate, fat, protein, vitamins, minerals and water. Carbohydrate, fat and protein yield metabolic energy; vitamins and minerals enable reactions but do not supply calories. Water is indispensable but energy-free. Alcohol supplies 7 kcal/g without being an essential nutrient.

REFERENCE	PRECISE MEANING	EXAM DISTINCTION
EAR	Intake estimated to meet the requirement of 50% of healthy people in a defined age/sex group	Useful for population assessment; not the goal that covers almost everybody.
RDA	Average daily intake sufficient for nearly all healthy individuals (about 97-98%) in the specified group	A planning goal over time, not an individual diagnostic cutoff or a treatment dose.
AI	Adequate intake set when evidence cannot establish an EAR/RDA	Not a deliberately lower RDA and not a guarantee that intake below it proves deficiency.
UL	Highest usual daily intake unlikely to pose adverse-effect risk for almost all healthy people in the group	Not a target. Absence of a UL does not establish unlimited safety.
EER	Predicted energy intake to maintain energy balance, allowing for growth/pregnancy/lactation where applicable	Unlike many nutrient RDAs it is not padded upward to cover everyone: persistent excess stores energy.

REFERENCE	PRECISE MEANING	EXAM DISTINCTION
AMDR	Acceptable range as a percentage of total daily energy	Adult carbohydrate 45-65%, fat 20-35%, protein 10-35%. Distinguish this from a lecturer's example distribution.

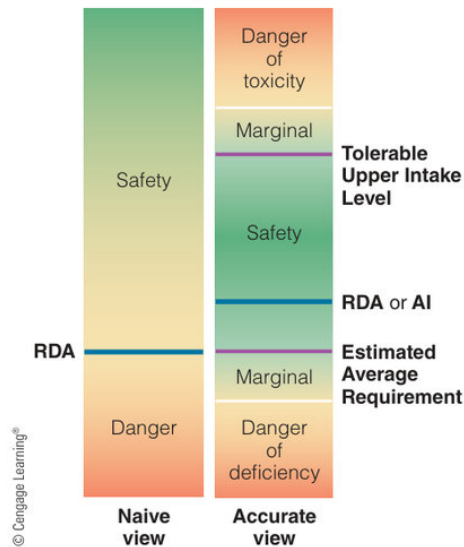


FIGURE 1-3 Naive versus Accurate View of Nutrient Intakes

The RDA or AI for a given nutrient represents a point that lies within a range of appropriate and reasonable intakes between toxicity and deficiency. Both of these recommendations are high enough to provide reserves in times of short-term dietary inadequacies, but not so high as to approach toxicity. Nutrient intakes above or below this range may be equally harmful.

Figure 1 Intake is a continuum: deficiency and excess can both harm. The diagram is conceptual, not a quantitative risk curve.

Source: DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016), PDF p. 32

- Nutrient density means nutrients relative to energy; energy density means kcal per gram. A food can be energy-dense and nutritionally valuable, such as nuts.
- Adequacy, balance, moderation and variety describe the overall pattern. Whole grains, vegetables, fruit and legumes contribute fiber and micronutrients, but preparation changes the fat and energy content.
- A portion is what someone serves or eats; a serving is a defined reference amount. They are not automatically equal. Compare labels using the same amount.
- Family custom and regional cuisine are cultural influences on food choice. Access, cost, appetite, religion and medical needs can modify those choices.

EXAM TRAP

Protein percentage in N1

The offered 10-15% option may reflect a conventional lecture distribution. The adult AMDR in the inspected book is 10-35%; none of that screenshot's options gives the full AMDR. Do not memorize 25-35% as the universal requirement.

CHECK YOURSELF**1 Does one day below an RDA diagnose deficiency?**

No. RDA is an intake planning standard, not a biochemical threshold. Assess usual intake, stores, absorption, losses, symptoms and appropriate tests.

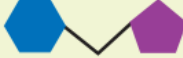
SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 29-32, 35-46 (PDF counter, not printed folio) · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q1-3, Q9-10; question evidence, not an official answer key · N - Nutrition .pdf - 28 general-nutrition scan pages - pp. 1, 20; question evidence, not an official answer key

03 / Carbohydrate, fiber and glycemic control

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

CLASS	EXAMPLES AND COMPOSITION	WHAT TO REMEMBER
Monosaccharides	Glucose, fructose, galactose	Single sugar units. Fructose is the expected sweetest common sugar among the F17 options.
Disaccharides	Sucrose = glucose + fructose; lactose = glucose + galactose; maltose = glucose + glucose	All three contain glucose. Lactose is not a monosaccharide.
Oligosaccharides	Short chains, commonly classified as 3-10 monosaccharide residues	Cutoffs vary by convention; F4 is not a secure single-best-answer item without the lecturer's definition.
Polysaccharides	Starch, glycogen and many dietary fibers	Starch is plant storage; glycogen is animal storage. Dietary fiber is a functional category and also includes noncarbohydrate lignin.

TABLE 2-1 The Major Sugars

Monosaccharides	Disaccharides
Glucose 	Sucrose (glucose + fructose) 
Fructose 	Lactose (glucose + galactose) 
Galactose  (found primarily as part of lactose)	Maltose (glucose + glucose) 

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Figure 2 The major sugars: use the paired shapes to reconstruct each disaccharide.

Source: DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016), PDF p. 58

Digestible starch is hydrolyzed to glucose. Humans cannot hydrolyze the beta linkages in cellulose. Some fibers are fermented by colonic microbes to short-chain fatty acids; others chiefly increase stool bulk. Soluble/insoluble is useful shorthand, but viscosity and fermentability better explain several effects.

FIBER BEHAVIOR	EXAMPLES	MECHANISM / CONSEQUENCE
Viscous soluble fiber	Pectin, oat/barley beta-glucan, psyllium	Can reduce bile-acid reabsorption and increase fecal bile-acid loss; the liver uses cholesterol to replenish bile acids. Can blunt post-meal glucose excursions.
Predominantly insoluble / bulking	Wheat bran and cellulose-rich foods	Adds bulk and may support stool transit. Increase intake with suitable fluid availability.
Fermentable substrates	Some soluble fibers, resistant starch and oligosaccharides	Support microbial fermentation; rapid increases can cause gas. Fiber types overlap rather than forming exclusive boxes.

- Insulin favors glucose uptake in responsive muscle/adipose tissue and storage; glucagon supports hepatic glucose release during fasting. Liver glycogen helps maintain blood glucose; muscle glycogen is used locally.
- Glycemic index compares the glucose response to a standardized available-carbohydrate amount; glycemic load also accounts for the carbohydrate quantity in a serving. Processing, ripeness, cooking, fat/protein and portion size matter.
- Lactose intolerance results from reduced lactase activity and can cause osmotic diarrhea and gas. It is not the same mechanism as immune-mediated cow's-milk protein allergy.
- Sugar alcohols such as xylitol are less cariogenic than sucrose but may cause gastrointestinal symptoms at high intake. Aspartame contributes phenylalanine and is unsuitable in PKU.

EXAM TRAP**The cholesterol-excretion question**

N6 and N18 repeat the same concept. Pectin is the best option among lactose, lignin, pectin and hemicellulose because of its viscous soluble-fiber behavior. Do not assume all fibers lower cholesterol to the same degree.

CHECK YOURSELF**1 Why can glucose remain necessary during prolonged fasting even when ketones rise?**

Red cells lack mitochondria and still require glucose; the brain increases ketone use but retains some glucose requirement. Fasting adaptation reduces, rather than abolishes, gluconeogenesis.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 57-63, 67-74, 77-79 (PDF counter, not printed folio) · F - nutrition final exam.pdf - 24 English screenshot pages - pp. 4-5, 17; question evidence, not an official answer key · N - Nutrition .pdf - 28 general-nutrition scan pages - pp. 6, 18; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q3-5; question evidence, not an official answer key

04 / Lipids, essential fatty acids and eicosanoids

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

MOLECULE	IDENTITY	EXAM TRAP
Triacylglycerol	Glycerol esterified to three fatty acids; principal dietary/storage fat	9 kcal/g, not 4 or 7.
Phospholipid	Amphipathic membrane/emulsifier molecule; lecithin means phosphatidylcholine	Linoleic acid is a fatty acid; inositol alone is not a phospholipid.
Cholesterol	Sterol in membranes and precursor of bile acids, steroid hormones and vitamin D	Not an essential fatty acid. Dietary cholesterol and blood LDL are not interchangeable measurements.
Saturated / mono- / polyunsaturated	Zero / one / multiple carbon-carbon double bonds	Cis/trans describes geometry of a double bond, not how many double bonds exist.

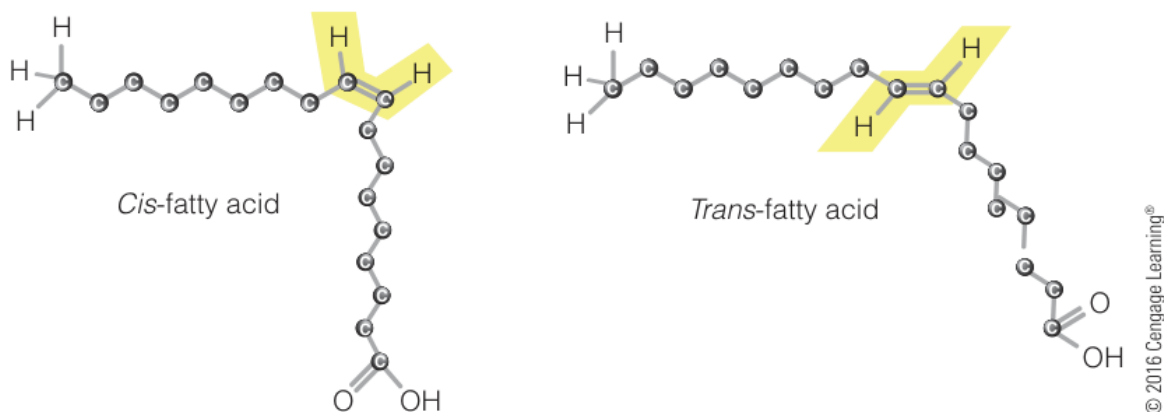


FIGURE 3-4 *Cis- and Trans-Fatty Acids Compared*

Figure 3 Cis and trans double-bond geometry affects molecular packing. A trans fatty acid remains chemically unsaturated.

Source: DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016), PDF p. 86

FATTY ACID	FAMILY	SOURCE / ROLE
Linoleic acid (18:2)	Omega-6; essential	Traditional high-linoleic sunflower/safflower oils; precursor for longer omega-6 derivatives. High-oleic cultivars are an important exception to the oil question.
Alpha-linolenic acid (18:3)	Omega-3; essential	Flax, chia, walnuts and some vegetable oils; limited conversion to EPA/DHA.
EPA / DHA	Long-chain omega-3	Seafood; membrane/signaling roles, with DHA important in neural/retinal tissue.
Oleic acid (18:1)	Omega-9; not an essential fatty acid	Abundant in olive oil and high-oleic varieties; not omega-3.

Fat carries fat-soluble vitamins, contributes to membranes and provides essential fatty acids. A very low-fat diet can jeopardize these functions; total fat percentage alone does not diagnose essential-fatty-acid deficiency because the actual linoleic and alpha-linolenic acid intakes matter.

LIPID MEDIATOR / CARRIER	CORE ROLE	DISTINGUISH
Thromboxane A₂	Promotes platelet aggregation and vasoconstriction	F12's "blood clotting" is a simplified correct association; prostacyclin opposes platelet aggregation.
Leukotrienes	Inflammatory signaling; some recruit leukocytes or constrict airways	Do not assign all leukocyte chemotaxis to thromboxane.
Chylomicrons / VLDL	Transport dietary / primarily liver-exported triglyceride	Chylomicrons initially enter lymph; VLDL is secreted into blood by liver.
LDL / HDL	LDL delivers cholesterol; HDL participates in reverse cholesterol transport	Clinical risk is not settled by an isolated "good/bad" label or a supplement intended simply to raise HDL.

CHECK YOURSELF

1 Why is replacing unsaturated oil with solid saturated fat a wrong dietary-guideline answer?

It reverses the usual favorable substitution. The replacement nutrient matters: choosing unsaturated fat in place of saturated fat is different from replacing fat with refined sugar.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 83-94, 97-99 (PDF counter, not printed folio) · F - nutrition final exam.pdf - 24 English screenshot pages - pp. 2, 12, 20; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q9, Q36-40; question evidence, not an official answer key

05 / Protein, amino acids and nitrogen balance

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

Amino acids contain an amino group, carboxyl group and a variable side chain. Glycine has hydrogen as its side chain and is the smallest standard amino acid. Proline's side chain loops back onto the amino nitrogen, restricting backbone flexibility. Phenylalanine and tryptophan also contain rings, so a question merely asking which amino acid is "cyclic" is imprecise.

- The nine essential amino acids are histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan and valine. Nonessential means synthesizable under usual conditions, not unimportant.
- Tyrosine becomes conditionally essential when conversion from phenylalanine is impaired in classic PKU. Dietary phenylalanine must be controlled, not indiscriminately eliminated; growth still requires essential amino acids.
- Tryptophan is a niacin precursor. Methionine metabolism produces homocysteine; remethylation uses folate/B12, while transsulfuration toward cysteine depends on B6.
- Protein has no inert dedicated storage depot analogous to adipose triglyceride. Body protein has structural or functional roles, and breakdown sacrifices tissue. Nitrogen is largely disposed of as urea after amino-acid catabolism.

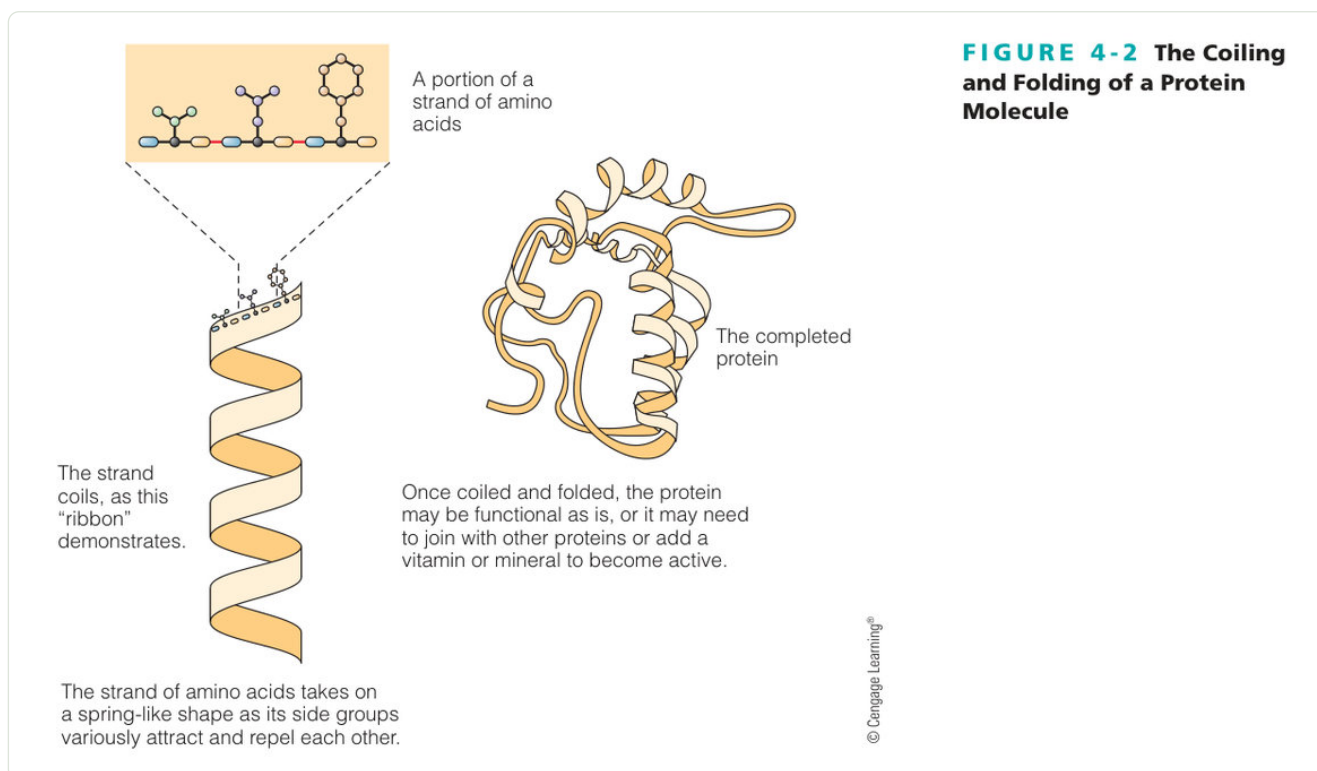


Figure 4 Sequence determines folding and function. Denaturation changes conformation without necessarily breaking every peptide bond.

Source: DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016), PDF p. 111

CONCEPT	MEANING	USEFUL EXAMPLE
Healthy adult RDA	0.8 g/kg/day as the conventional reference	A healthy 70-kg adult: 56 g/day. Illness, growth, pregnancy and other contexts can alter the target.
Positive nitrogen balance	Retention exceeds nitrogen loss	Growth, pregnancy and tissue recovery with adequate nutrition.
Negative nitrogen balance	Loss exceeds intake/retention	Starvation, substantial trauma or catabolic illness; severity and treatment affect the direction.
Limiting amino acid	Essential amino acid least adequate relative to synthesis needs	Legumes and grains can complement each other across an adequate varied diet; every meal need not be a rigid matched pair.
Protein quality	Digestibility plus indispensable amino-acid pattern	Soy is a useful high-quality plant protein; gelatin is not nutritionally complete.

EXAM TRAP

Fasting is not an on/off fuel queue

Protein contributes to gluconeogenesis before every carbohydrate and fat store has been exhausted. During prolonged starvation ketone use helps spare protein; terminal fat depletion accelerates destructive proteolysis. D5 describes an endpoint, not the full time course.

CHECK YOURSELF**1 Why do carbohydrate and fat availability change the effective use of dietary protein?**

If energy is inadequate, amino-acid carbon skeletons are oxidized for fuel rather than all being available for synthesis; adequate energy has a protein-sparing effect.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 109-120, 487-490 (PDF counter, not printed folio) · F - nutrition final exam.pdf - 24 English screenshot pages - pp. 16, 22; question evidence, not an official answer key · N - Nutrition .pdf - 28 general-nutrition scan pages - pp. 3, 17, 24; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q5, Q31-35; question evidence, not an official answer key

06 / Digestion and absorption: predict the deficiency

Source basis Book extension supporting the past-paper mechanisms

SITE / PROCESS	SUBSTRATES AND PRODUCTS	CONSEQUENCE OF FAILURE
Mouth and stomach	Salivary amylase starts starch digestion. Gastric acid denatures proteins and supports pepsin activation.	Chewing/swallowing problems reduce intake; achlorhydria impairs release of food-bound B12.
Pancreatic enzymes	Amylase, lipase and proteases act in the small intestine; bicarbonate neutralizes acid	Pancreatic insufficiency can cause maldigestion, especially fat malabsorption.
Bile and micelles	Bile salts emulsify fat and facilitate delivery of lipid digestion products to enterocytes	Cholestasis or substantial bile-salt loss can compromise fat and A/D/E/K absorption. Bile is not a digestive enzyme.
Brush border	Disaccharidases and peptidases complete digestion	Lactase deficiency leaves an osmotic and fermentable substrate.
Portal versus lymphatic route	Monosaccharides/amino acids mainly enter portal blood; long-chain fat is re-esterified and packaged in chylomicrons entering lymph	Lymph is not the usual initial route for absorbed glucose.
B12 pathway	Gastric release, binding-protein processing, intrinsic-factor complex and terminal-ileal uptake	Gastric or ileal disease may cause deficiency despite adequate intake.

Intestinal folds, villi and microvilli increase surface area. Nutrient intake is therefore not equal to nutrient availability: digestion, absorption, binding, losses, drug interactions and disease can intervene. This is why a food diary cannot exclude deficiency in a patient with malabsorption.

CHECK YOURSELF**1 A person has steatorrhea plus easy bruising and bone pain. Which nutrient families need attention?**

Fat-soluble vitamins, especially K for coagulation and D for mineralization, alongside energy, essential fatty acids and other malabsorbed nutrients. This is a mechanistic differential, not a diagnosis from two symptoms.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 130-144, 231 (PDF counter, not printed folio)

07 / Energy calculations, BMR and fasting

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

QUANTITY	FORMULA / APPROXIMATION	CHECK
Food energy	$4 \times \text{carbohydrate g} + 4 \times \text{protein g} + 9 \times \text{fat g}$; add $7 \times \text{alcohol g}$ if present	Use grams, then kcal. Do not apply 9 to all nutrients.
Percent energy	$\text{Nutrient kcal} / \text{total kcal} \times 100$	The denominator is total energy, not total food weight.
Protein grams from a percentage	$\text{Total kcal} \times \text{protein fraction} / 4$	15% of 2,000 kcal = 300 kcal = 75 g protein.
TEF	Approximately 10% of total energy expenditure for a mixed diet	A teaching estimate, not a universal fixed percentage; macronutrient mix changes it.
TEE	Resting/basal expenditure + physical activity + thermic effect of food (plus growth etc. when applicable)	Basal expenditure is usually the largest component in sedentary adults; activity is highly variable.

Basal metabolism supports heartbeat, respiration, membrane transport and temperature maintenance. Walking/running are activity expenditure. BMR is measured under standardized awake, resting, postabsorptive and thermoneutral conditions. RMR uses less restrictive conditions. Lean mass, growth, fever and thyroid activity influence resting expenditure; loss of lean mass and adaptive responses to restriction can lower it.

EXAM TRAP

Measurement versus prediction

Direct calorimetry measures heat; indirect calorimetry estimates expenditure from respiratory gas exchange. Equations predict expenditure and are less individualized than an appropriate measurement. N26's selected formula option should not be accepted as the most accurate method. If the exam distinguishes practical measured REE, indirect calorimetry is commonly intended; a literal heat-measurement question can mean direct calorimetry.

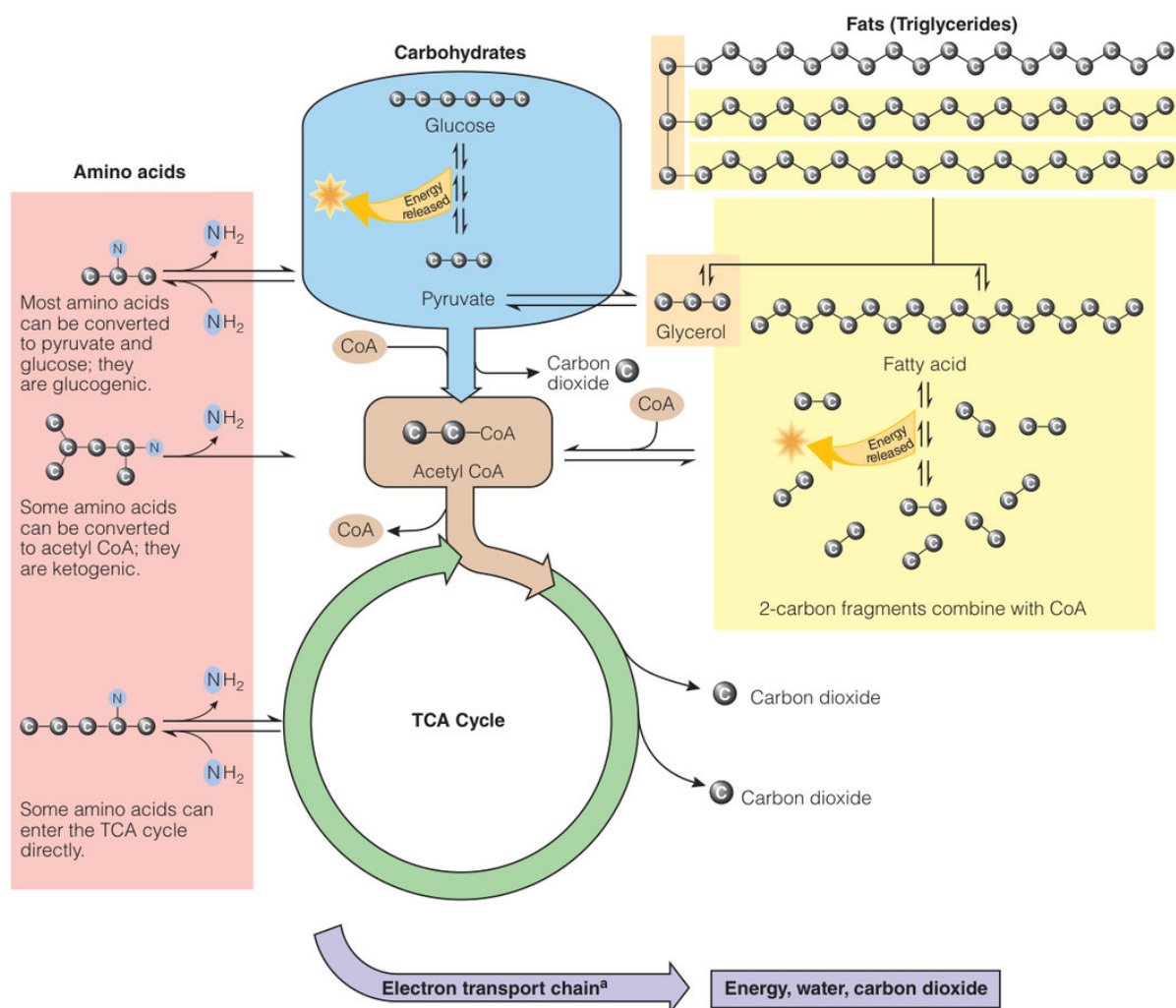


FIGURE 6-3 The Central Pathways of Energy Metabolism

Figure 5 Trace carbohydrate, amino-acid and fat carbon into the central pathways. The figure is a teaching overview, not an exhaustive pathway map.

Source: DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016), PDF p. 161

Insulin predominates after feeding and favors storage. During fasting, glycogenolysis, lipolysis and gluconeogenesis overlap. Alanine, lactate and glycerol can support gluconeogenesis; even-chain fatty-acid acetyl-CoA does not provide net glucose in humans. Ketones become a larger fuel in prolonged fasting. Do not confuse nutritional ketosis with the dangerous physiology of diabetic ketoacidosis.

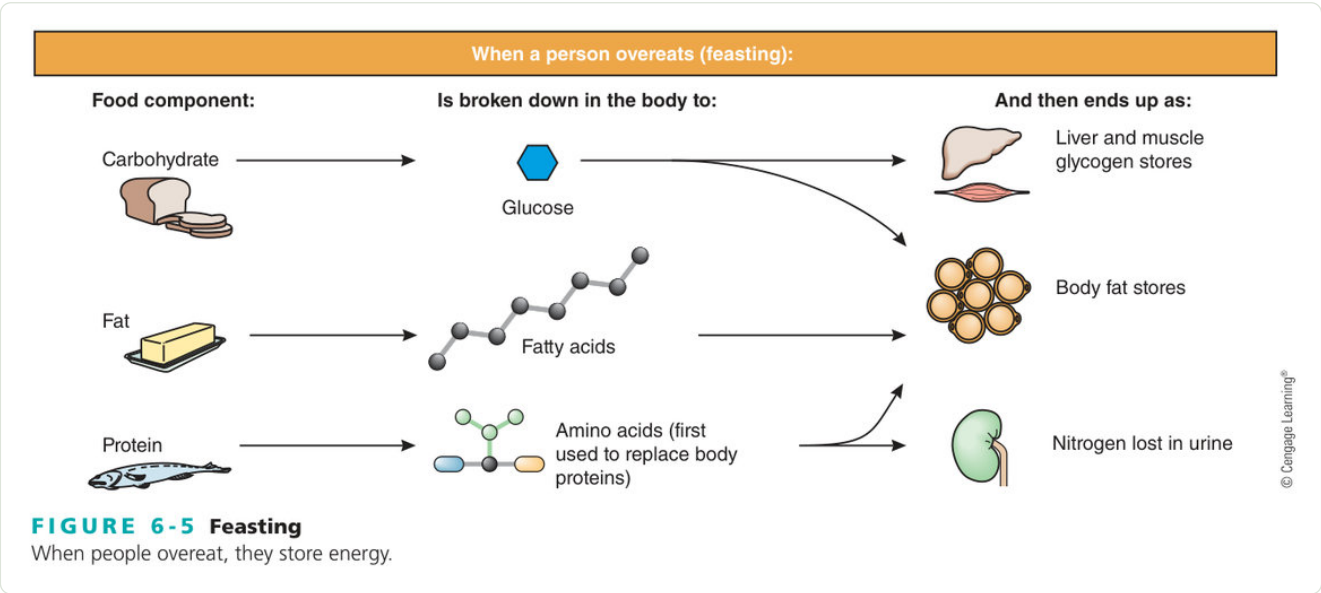


Figure 6 Excess absorbed energy can enlarge stores. Nitrogen must be disposed of when amino acids are used as fuel.
Source: DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016), PDF p. 163

CHECK YOURSELF

- 1 What does 7 g fat plus 7 g protein provide?
 $7 \times 9 + 7 \times 4 = 63 + 28 = 91 \text{ kcal}$. The fat supplies $63/91 \times 100 = 69.2\%$ of this energy.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 29-30, 158-170 (PDF counter, not printed folio) · F - nutrition final exam.pdf - 24 English screenshot pages - pp. 2, 8; question evidence, not an official answer key · N - Nutrition .pdf - 28 general-nutrition scan pages - pp. 7, 25-26; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q5-7, Q37; question evidence, not an official answer key

08 / Assessment, BMI and body composition

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

Start with the question being asked: dietary intake, nutritional status, body composition and disease risk are related but different. A single laboratory result or BMI cannot answer all four.

DOMAIN	METHODS	LIMITS
Anthropometric	Weight trend, height, BMI, waist, age-appropriate growth measures	Fluid retention may conceal tissue loss; BMI cannot separate muscle from fat or identify fat distribution.
Biochemical	Context-appropriate blood/urine tests	Inflammation, renal/hepatic disease, hydration and recent intake affect many values.
Clinical / functional	History, examination, muscle/fat loss, edema, oral health, strength and function	Signs are often nonspecific; combine findings rather than diagnosing from one symptom.
Dietary	24-hour recall, food record, food-frequency history	Recall bias, under-reporting, portion error and atypical days are common.

DOMAIN	METHODS	LIMITS
Indirect community indicators	Morbidity, mortality, food availability and social/economic measures	Population context, not an individual nutrient assay.

BMI = mass in kg / height in m squared. Conventional adult categories: underweight <18.5; reference healthy range 18.5-24.9; overweight 25.0-29.9; obesity ≥30. These adult boundaries are not automatically applied to children, whose BMI is interpreted for age and sex. Waist circumference reflects central adiposity; waist-to-hip ratio uses waist / hip measured in the same units. Central adiposity can confer risk despite a BMI in the reference range.

EXAM TRAP**Albumin is not a protein-intake meter**

N2's albumin option reflects a familiar older assessment shortcut. Albumin and prealbumin are strongly affected by inflammation and illness and should not be used alone to quantify intake, muscle mass or malnutrition. Use intake history, trajectory, examination and clinical context.

CHECK YOURSELF**1 An adult is 81 kg and 1.80 m tall. What is BMI, and what remains unknown?**

$81 / 3.24 = 25.0 \text{ kg/m}^2$. This lies at the conventional overweight threshold, but fat distribution, muscle mass, edema, recent weight change and individual risk remain unknown.

Oral-paper crossover: weight change is not one mechanism

CONTEXT	WHY IT APPEARS IN A WEIGHT QUESTION	LIMIT
Cushing syndrome / glucocorticoid exposure	Can promote weight gain and altered fat distribution; muscle weakness may coexist	An increase in weight does not imply more muscle.
Hypothyroidism	Reduced thyroid function can accompany weight gain	Common symptoms alone do not establish the diagnosis.
Adult growth hormone deficiency	Can shift composition toward more fat and less lean tissue	Not equivalent to growth-hormone excess, and not a routine explanation for all obesity.
Prednisolone	Can increase appetite and cause fluid retention and weight gain	Dose, duration and individual context matter; do not stop prescribed steroids from an exam note.
Sjögren syndrome	The relevant association is salivary dysfunction and dry mouth	It is the intended exception in O1, not a rule that an affected person cannot gain weight.

CHECK YOURSELF**1 Why can medication-related weight gain coexist with loss of functional muscle?**

Weight sums fat, lean tissue, fluid and other compartments. A process can increase fat or retained fluid while impairing muscle. Reassess function and body composition instead of interpreting the scale as a nutrition verdict.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 169-174, 419-428 (PDF counter, not printed folio) · N - Nutrition .pdf - 28 general-nutrition scan pages - pp. 2, 4, 11, 16, 26; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q8; question evidence, not an official answer key · ASPEN - Visceral proteins as nutrition markers - Checked 14 September 2026 - <https://aspenjournals.onlinelibrary.wiley.com/doi/10.1002/ncp.10588> · NIDDK - Factors Affecting Weight and Health - Medicines and health conditions; checked 14 September 2026 - <https://www.niddk.nih.gov/health-information/weight-management/adult-overweight-obesity/factors-affecting-weight-health> · Endocrine Society - Adult Growth Hormone Deficiency Guideline - Adult body composition; checked 14 September 2026 - <https://www.endocrine.org/clinical-practice-guidelines/adult-growth-hormone-deficiency> · NHS - Prednisolone side effects - Weight gain; checked 14 September 2026 - <https://www.nhs.uk/medicines/prednisolone/side-effects-of-prednisolone-tablets-and-liquid/>

09 / Protein-energy malnutrition and refeeding

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

PATTERN	CORE FEATURES	AVOID THE SHORTCUT
Marasmus / severe wasting	Marked depletion of fat and muscle from inadequate energy/nutrients; nonedematous wasting pattern	Not “protein is normal.” Multiple deficiencies and infection can coexist.
Kwashiorkor / nutritional edema	Bilateral pitting edema, skin/hair changes, fatty liver and impaired defenses may occur	The old isolated-protein-deficiency story is incomplete; inflammatory, oxidative, infectious and metabolic factors contribute.
Wasting / stunting	Low weight for height / low height for age in children	They describe different growth deficits and can coexist.
Sarcopenia	Loss of muscle strength/function with reduced muscle quantity/quality	Can occur with normal or high BMI; weight alone misses it.

Illness can reduce intake, increase needs or losses, impair absorption and accelerate catabolism at the same time. Edema can increase measured weight while tissue stores fall. Severe malnutrition therefore requires assessment beyond a simple calorie calculation.

CLINICAL LINK**Refeeding is a physiology problem**

After prolonged undernutrition, carbohydrate-driven insulin shifts phosphate, potassium and magnesium into cells and increases thiamin demand. Electrolyte and fluid complications can be dangerous. High-risk patients need supervised refeeding and monitoring, not an abrupt unsupervised “catch-up” diet.

CHECK YOURSELF**1 Why might a malnourished child look swollen rather than thin?**

Edema can mask muscle/fat loss. Nutritional edema is an important severe-malnutrition phenotype; it must not be mistaken for adequate nutritional reserves.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 116-118, 424-426, 480 (PDF counter, not printed folio) · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q32, Q35; question evidence, not an official answer key

10 / Vitamin A: vision, epithelium and toxicity

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

FORM / FUNCTION	MECHANISM	CLUE
Retinal	11-cis retinal participates in visual pigment; light causes isomerization	Night blindness is an early classic deficiency symptom.
Retinoic acid	Regulates gene expression and epithelial differentiation	Do not select it as the direct visual-pigment chromophore.
Retinol / retinyl esters	Transport/storage-related forms	Retinol-binding protein is a carrier, not a vitamin-A chemical form responsible for vision.
Provitamin A carotenoids	Plant precursors, including beta-carotene	Deep green/orange produce can contribute vitamin A equivalents; not all carotenoids have provitamin activity.

- Deficiency can cause impaired dark adaptation, conjunctival/corneal dryness, Bitot spots and severe corneal injury; epithelial and immune functions are also affected.
- Liver is particularly rich in preformed vitamin A. Carrots, pumpkin and dark green leaves provide carotenoid precursors. Serving size and preparation matter in food comparisons.
- Excess preformed vitamin A can cause toxicity and is teratogenic. This explains the past-paper warning about excessive liver intake during pregnancy; it does not mean all vitamin-A-containing foods should be avoided.
- Adult reference intakes are 900 micrograms RAE/day for men and 700 for women; units are retinol activity equivalents, not automatically the same as micrograms of every carotenoid.

CHECK YOURSELF**1 A stem pairs night blindness with a request for the active visual form. Give both answers.**

Vitamin A deficiency; retinal is the visual-cycle form. Retinoic acid primarily mediates gene-regulatory effects.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 212-218 (PDF counter, not printed folio) · F - nutrition final exam.pdf - 24 English screenshot pages - pp. 7, 24; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q15, Q18, Q25; question evidence, not an official answer key

11 / Vitamin D, calcium and mineralization

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

Skin 7-dehydrocholesterol is converted through UVB-dependent steps to vitamin D₃. Liver 25-hydroxylation produces 25-hydroxyvitamin D (calcidiol); primarily renal 1-alpha-hydroxylation produces 1,25-dihydroxyvitamin D (calcitriol). The active hormone supports intestinal calcium and phosphate absorption and normal mineralization.

CONDITION	WHAT FAILS	TYPICAL DISTINCTION
Rickets	Mineralization in the growing skeleton / growth plates	Bowed legs, widened growth regions and rachitic rosary at costochondral junctions. Pigeon-chest deformity may appear but is not specific.
Osteomalacia	Mineralization of adult osteoid	Bone pain and fragility; not merely low total bone mass.
Osteoporosis	Reduced bone mass and microarchitectural deterioration	Remaining matrix is generally normally mineralized; fracture risk depends on several factors.
Vitamin D excess	Excessive calcium absorption and dysregulated calcium balance	Hypercalcemia and renal injury risk, principally from excessive supplements rather than ordinary sun exposure.

Serum 25(OH)D is the usual status marker; calcitriol is tightly regulated and can be maintained despite low stores. Adult DRI: 600 IU (15 micrograms) daily through age 70, including pregnancy/lactation; 800 IU (20 micrograms) after 70. Infants have a 400 IU (10 micrograms) adequate intake. One microgram vitamin D = 40 IU. These are intake references, not deficiency treatment prescriptions.

CHECK YOURSELF

1 Why can normal serum calcium coexist with inadequate calcium intake?

Homeostatic regulation can maintain blood calcium partly at the expense of skeletal mineral. A normal concentration does not prove adequate bone stores or dietary intake.

Past-paper distinction: activation is not every renal hydroxylation

STEP	PRODUCT / ROLE	ANSWER THE ACTUAL WORDING
Liver: 25-hydroxylation	Vitamin D becomes 25(OH)D, the principal circulating status marker	This is not the usual renal activation step.
Kidney: CYP27B1 / 1-alpha-hydroxylation	25(OH)D becomes 1,25(OH) ₂ D (calcitriol)	Correct when the question specifies activation.
Kidney and other target tissues: CYP24A1	24-hydroxylation initiates pathways that limit vitamin D activity	This also occurs in the kidney; it cannot be excluded merely by naming that organ.

EXAM TRAP

F21 remains ambiguous, not a certified corrected MCQ

The original stem asks about hydroxylation in the kidney without clearly restricting it to activation. Both renal 1-alpha-hydroxylation and renal 24-hydroxylation are real. Learn the pathway distinction; do not treat the marked option or a forced replacement key as official. MED25 therefore retains F21 for source review rather than automatic scoring.

CHECK YOURSELF

1 Why can '24-hydroxylation occurs in kidney' and 'renal activation is 1-alpha-hydroxylation' both be true?

They describe different enzymes and purposes. CYP27B1 produces calcitriol; CYP24A1 regulates turnover through hydroxylated products. Organ location alone does not identify the activation pathway.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 218-221, 254-257 (PDF counter, not printed folio) · F - nutrition final exam.pdf - 24 English screenshot pages - pp. 9, 21, 23; question evidence, not an official answer key · N - Nutrition .pdf - 28 general-nutrition scan pages - pp. 5, 8; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q12; question evidence, not an official answer key · NIH ODS - VitaminD - Checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/VitaminD-HealthProfessional/> · National Academies - Dietary Reference Intakes for Calcium and Vitamin D: Overview of Vitamin D - Vitamin D metabolism and CYP24A1; checked 14 September 2026 - <https://www.ncbi.nlm.nih.gov/books/NBK56061/>

12 / Vitamins E and K: antioxidant versus clotting roles

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

VITAMIN	CORE ROLE	DEFICIENCY / SAFETY
E (alpha-tocopherol)	Lipid-phase antioxidant that interrupts lipid-peroxidation chain reactions	Severe deficiency may cause neurological and hemolytic problems, especially with fat malabsorption. High-dose supplements are not a general disease-prevention strategy.
K	Cofactor for gamma-carboxylation of specific proteins, including clotting factors II, VII, IX, X and proteins C/S	Deficiency impairs coagulation. Newborn prophylaxis is a separate clinical protocol, not simply an adult dietary RDA.

Selenium-dependent glutathione peroxidases remove peroxides; vitamin E limits propagation of membrane lipid oxidation. This complementary antioxidant relationship explains F6. It does not imply that giving large doses of both improves outcomes in a nutrient-replete person.

- Plant oils, nuts and seeds are useful vitamin E sources. Green leafy vegetables are important vitamin K sources.
- Fat malabsorption threatens both. Vitamin E is not the principal clotting vitamin; K is not the answer to the classic antioxidant question.
- With vitamin-K antagonists such as warfarin, consistent intake and clinical monitoring matter; indiscriminate elimination of vegetables is not the lesson.

CHECK YOURSELF

1 Which nutrient pairs with vitamin E in peroxide defense, and which vitamin supports clotting-factor maturation?

Selenium; vitamin K. Keep antioxidant cooperation separate from gamma-carboxylation.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 221-224, 267 (PDF counter, not printed folio) · F - nutrition final exam.pdf - 24 English screenshot pages - pp. 6, 10; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q17; question evidence, not an official answer key

13 / B vitamins: enzymes and deficiency patterns

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

VITAMIN / COENZYME	KEY BIOCHEMICAL WORK	DEFICIENCY / EXAM CLUE
B1 thiamin / TPP	Oxidative decarboxylation and transketolase reactions	Beriberi: peripheral neuropathy and/or cardiac manifestations; Wernicke encephalopathy risk in severe deficiency, often with alcohol-use disorder.
B2 riboflavin / FMN, FAD	Redox reactions	Angular mouth lesions, glossitis and ocular symptoms can occur. The classic photophobia association in D13 is riboflavin, but is not diagnostic alone.
B3 niacin / NAD, NADP	Electron transfer in energy metabolism and synthesis	Pellagra: dermatitis, diarrhea, dementia and, untreated, death. Tryptophan contributes to niacin synthesis.
B5 pantothenate / CoA	Acyl-group transfer	Deficiency is uncommon in ordinary mixed diets; widespread food distribution.
B7 biotin	Carboxylation reactions	Raw egg-white avidin can bind biotin. Cooking denatures avidin. Deficiency is uncommon.

B vitamins assist energy metabolism; they do not themselves yield calories. Severe thiamin deficiency particularly threatens high-demand nervous and cardiac tissue. Classical “dry” beriberi emphasizes neuropathy and “wet” beriberi cardiovascular involvement. The Wernicke pattern includes mental-status change, ocular dysfunction and ataxia, although the full triad need not be present.

- The conventional niacin equivalent includes approximately 60 mg tryptophan per 1 mg niacin equivalent; conversion is affected by nutritional/metabolic conditions.
- Riboflavin is light-sensitive; thiamin and vitamin C are vulnerable to processing/heat-related losses. Protecting foods from prolonged heat, light and unnecessary discarded cooking water preserves susceptible nutrients.
- A symptom cluster is not pathognomonic. D16’s anorexia, fatigue, arrhythmia and distal tingling suggests thiamin among its choices, but a real patient requires a broader assessment.

CHECK YOURSELF**1 How can an apparently carbohydrate-rich diet coexist with impaired carbohydrate oxidation?**

If a required cofactor such as thiamin is deficient, pyruvate oxidation is impaired despite abundant substrate. More sugar does not repair the cofactor deficit.

Past-paper distinction: four thiamin-deficiency presentations

PRESENTATION	DOMINANT PATTERN	DISTINGUISHING POINT
Dry beriberi	Peripheral neuropathy and muscle weakness	Not the folate/B12 megaloblastic-anemia mechanism.
Wet beriberi	Cardiovascular dysfunction, potentially heart failure and edema	Connect cardiac energy demand to thiamin-dependent metabolism.
Wernicke encephalopathy	Acute mental-status change, gait ataxia and eye-movement abnormalities	The entire classic triad need not be present. Severe deficiency is urgent.
Korsakoff syndrome	Persistent severe memory impairment; confabulation can occur	A chronic amnesic syndrome associated with thiamin deficiency, not a synonym for every acute confused state.

Infants receiving milk from a severely thiamin-deficient mother may themselves become deficient. O10's marked 2-5-month interval is an exam-specific age claim, not a biological boundary. The actionable learning point is maternal supply plus the infant's limited reserves and high metabolic demand. Wernicke-Korsakoff is associated with alcohol-use disorder but is not restricted to it.

CHECK YOURSELF**1 A question names Wernicke-Korsakoff; another names beriberi. Must the vitamin differ?**

No. Both test thiamin/B1, but the phenotype differs. Explain acute central neurological dysfunction, persistent memory injury, peripheral neuropathy and cardiac involvement separately rather than memorizing unrelated vitamin pairs.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 225-228, 235-236 (PDF counter, not printed folio) · F - nutrition final exam.pdf - 24 English screenshot pages - p. 15; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q13, Q16, Q33; question evidence, not an official answer key · NIH ODS - Thiamin: Health Professional Fact Sheet - Deficiency and Wernicke-Korsakoff sections; checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/Thiamin-HealthProfessional/> · NIAAA - Wernicke-Korsakoff Syndrome - Acute/chronic syndromes and confabulation; checked 15 September 2026 - <https://www.niaaa.nih.gov/publications/brochures-and-fact-sheets/wernicke-korsakoff-syndrome> · WHO - Thiamine deficiency and its prevention and control in major emergencies - Maternal/infant deficiency relationship; checked 15 September 2026 - https://iris.who.int/bitstream/handle/10665/66139/WHO_NHD_99.13.pdf

14 / B6, folate and B12: connect the pathways

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

NUTRIENT	CORE MECHANISM	HIGH-YIELD DISTINCTION
B6 / PLP	Amino-acid metabolism, neurotransmitter and heme synthesis; glycogen phosphorylase cofactor	Deficiency can cause dermatitis, neurological problems and anemia; excessive chronic supplementation can itself cause neuropathy.
Folate / THF derivatives	One-carbon transfer in nucleotide synthesis and methylation metabolism	Rapidly dividing tissues are vulnerable. Deficiency causes megaloblastic anemia; adequate periconceptional folate reduces neural-tube-defect risk.
B12 / cobalamin	Methionine synthase and methylmalonyl-CoA mutase reactions	Megaloblastic anemia plus potentially irreversible neurological injury. Methylmalonic acid rises in deficiency, with interpretation affected by renal function.

Homocysteine has two key exits: remethylation to methionine requires folate/B12; transsulfuration toward cysteine requires B6. Homocysteine elevation is therefore not specific for one vitamin. B12 deficiency can trap folate in a form unavailable for normal nucleotide synthesis. Folate can improve the anemia while allowing untreated B12-related neurological injury to progress.

REFERENCE AMOUNT	MEANING	DO NOT CONFUSE
B12: 2.4 micrograms/day adult	2.6 in pregnancy; 2.8 in lactation	F11 offers 2 micrograms as a rounded/older option, not the exact current adult DRI. Milligrams would be a thousand-fold different unit.
Folate: 400 micrograms DFE/day adult	600 in pregnancy; 500 in lactation	DFE includes different bioavailability of food folate and folic acid.
Periconceptional folic acid: 400 micrograms/day	From supplements/fortified foods in addition to a varied folate-containing diet, under the cited population guidance	Not the same quantity as total pregnancy folate DFE; high-risk regimens require clinician direction.

Folate-rich foods include legumes and leafy vegetables; fortification varies by food and country. Reliable B12 sources include animal foods or appropriately fortified foods/supplements. A vegan diet requires a reliable B12 source. Gastric disease, ileal disease and some drugs can cause deficiency even with dietary B12.

EXAM TRAP

Drug interactions need a named drug

Folate is the marked choice for D14's "most vulnerable vitamin" claim. Learn specific mechanisms instead: antifolate drugs affect folate pathways; some anticonvulsants interact with folate; isoniazid is associated with B6 depletion; metformin can reduce B12 status. There is no universally useful ranking of all vitamins without specifying the medication and context.

CHECK YOURSELF**1 Anemia improves with folate, but numbness progresses. What danger was missed?**

Possible B12 deficiency with continuing neurological injury. Correction of the blood picture is not proof the underlying deficiency has been safely treated.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 228-232, 316-317 (PDF counter, not printed folio) · F - nutrition final exam.pdf - 24 English screenshot pages - pp. 11, 18; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q14, Q21, Q24, Q34; question evidence, not an official answer key · NIH ODS - VitaminB12 - Checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/VitaminB12-HealthProfessional/> · NIH ODS - Folate - Checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/Folate-HealthProfessional/>

15 / Vitamin C, connective tissue and antioxidant logic

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

Vitamin C supports collagen hydroxylation, carnitine synthesis and catecholamine-related reactions, and acts as a water-phase antioxidant. It enhances nonheme iron absorption when consumed with an iron-containing meal. These mechanisms connect bleeding gums, poor wound healing and an iron-absorption question to the same nutrient.

- Scurvy reflects inadequate vitamin C: fragile connective tissue, petechiae/bruising, gingival abnormalities, poor healing and musculoskeletal pain can occur. Anemia can have several contributors.
- Citrus, peppers, berries and many vegetables are useful sources. Vitamin C is vulnerable to oxygen, storage and cooking losses; a vitamin-C-rich food is not necessarily useful if processing has removed much of it.
- Adult reference intakes are 90 mg/day in men and 75 mg/day in women, with an additional 35 mg/day for smokers in the U.S. DRI framework. These are not megadose targets.
- Water-soluble does not mean unlimited safety. High intake may cause gastrointestinal problems; disease context can change risk.

CHECK THE SOURCE**Do not diagnose neonatal bleeding from a maternal-vitamin story**

O34 implies neonatal gum bleeding proves excessive maternal vitamin C. That is not a safe diagnostic rule. Neonatal bleeding has serious alternative causes, and the historical “rebound scurvy” assertion should not be taught as an established answer without the lecturer’s evidence.

CHECK YOURSELF**1 Why can the same dietary change help both collagen maintenance and iron absorption?**

Adding a vitamin-C-rich food supplies a collagen-reaction cofactor and enhances absorption of nonheme iron consumed in the same meal. This does not replace investigation of blood loss or malabsorption.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 232-234, 513-514 (PDF counter, not printed folio) · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q22; question evidence, not an official answer key · O - Taghziyeh.pdf - Nutrition in Oral Health paper - Q18, Q29, Q34; question evidence, not an official answer key

16 / Iron: intake, transport, stores and anemia

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

Iron participates in hemoglobin, myoglobin and enzymes. Transferrin transports it; ferritin stores it. Hepcidin controls availability by reducing ferroportin-mediated export from enterocytes and iron-recycling/storage cells. Inflammation can increase hepcidin and restrict iron availability despite retained stores.

STATE	TYPICAL PATTERN / MECHANISM	INTERPRETIVE TRAP
Depleted stores	Ferritin falls before hemoglobin necessarily falls	Iron deficiency can exist without anemia.
Iron-deficient erythropoiesis / anemia	Reduced iron supply, then reduced hemoglobin; often microcytic/hypochromic with progression	Not every microcytic anemia is iron deficiency.
Inflammatory anemia	Iron sequestration and altered erythropoiesis; ferritin may be normal/high	Normal or high ferritin during inflammation does not reliably exclude concurrent iron deficiency.
Thalassemia / sickle-cell disease	Inherited globin disorders	Neither is caused by iron deficiency; iron should not be assumed helpful without evidence of a deficiency.

GROUP	IRON RDA	REASON FOR DISTINCTION
Adult men / adults over 50	8 mg/day	Lower usual requirements than menstruating young women.
Women 19-50, nonpregnant	18 mg/day	Menstrual loss affects needs.
Pregnancy	27 mg/day	Maternal and fetal demands increase.
Lactation age 19-50	9 mg/day	Different life-stage physiology; not the pregnancy value.

- Heme iron is generally better absorbed. Nonheme iron absorption depends more on the meal; vitamin C helps, while tea polyphenols and phytate can inhibit it.
- Blood loss, growth, inadequate intake and malabsorption are major causes. Chronic infection classically causes inflammatory anemia, although it can coexist with iron deficiency.
- Dietary prevention emphasizes suitable iron sources and meal composition. Do not infer that everyone should take iron tablets or stop all dairy products.
- Pregnancy plasma volume expands proportionally more than red-cell mass, causing physiological hemodilution; this does not remove the need to assess iron deficiency.

EXAM TRAP

A dangerous marked-answer error

N27 selects chronic occult fecal bleeding as the option that would not cause iron-deficiency anemia. That is wrong: sustained blood loss is a classic cause. Chronic infection is the better distinction among those options because it more characteristically produces anemia of inflammation.

CHECK YOURSELF**1 Low hemoglobin in pregnancy: physiological, iron-deficient, or both?**

All are possible. Use gestational context and the clinical/laboratory assessment; hemodilution and iron depletion can coexist.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 261-265, 318-319 (PDF counter, not printed folio) · F - nutrition final exam.pdf - 24 English screenshot pages - p. 1; question evidence, not an official answer key · N - Nutrition .pdf - 28 general-nutrition scan pages - pp. 15, 22, 27-28; question evidence, not an official answer key · O - Taghziyeh.pdf - Nutrition in Oral Health paper - Q5, Q26; question evidence, not an official answer key · NIH ODS - Iron - Checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/Iron-HealthProfessional/>

17 / Water, major minerals and trace elements

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

NUTRIENT	MECHANISM AND CLASSIC CLUE	SOURCE / CAUTION
Sodium / chloride	Major extracellular ions; volume, acid-base and nerve/muscle physiology	Salt and processed foods; excess sodium can contribute to hypertension. Sodium concentration is also strongly about water balance.
Potassium	Principal intracellular cation; membrane excitability	Legumes, potatoes, fruit and vegetables. Apricot/banana are useful sources; cucumber is lower among D20 choices. Hyperkalemia risk changes advice in kidney disease.
Calcium	Skeleton, contraction, signaling and coagulation	Dairy, suitable fortified foods, and edible-bone fish. Food comparisons need a serving size.
Phosphorus	Hydroxyapatite, ATP, phospholipids and DNA/RNA phosphate backbone	F14 asks phosphorus, not iron or calcium.
Magnesium	Many enzyme reactions and neuromuscular function	Nuts, legumes, whole grains and greens; severe depletion can accompany potassium/calcium disturbances.
Zinc	Enzymes, growth, immunity, taste and wound repair	Meat/shellfish, dairy, legumes/nuts; dysgeusia, alopecia and delayed healing are classic. Excess zinc can cause copper deficiency.
Selenium	Selenoproteins including glutathione peroxidases and thyroid-hormone metabolism enzymes	Keshan cardiomyopathy association; content varies by soil. Excess can damage hair/nails and cause systemic toxicity.
Iodine	Thyroid hormone synthesis	Iodized salt and appropriate foods; deficiency can cause goiter and harm neurodevelopment. Excess can also disturb thyroid function.

NUTRIENT	MECHANISM AND CLASSIC CLUE	SOURCE / CAUTION
Copper	Enzyme cofactor in iron handling and connective tissue; ceruloplasmin is a major circulating copper-containing protein	Organ meats, shellfish, nuts/seeds. Ceruloplasmin is not the body's principal storage organ.
Fluoride	Promotes enamel resistance/remineralization in appropriate exposure	Excess during enamel development causes fluorosis; dental products are not intended to be swallowed.
Manganese / molybdenum	Enzyme cofactors	Distinguish trace amounts required from a recommendation to supplement.

Water balance depends on intake, renal handling and other losses. Thirst may be blunted in older adults; dehydration can impair function and cognition. Excess water relative to excretory capacity can cause hyponatremia. Thus “more water is always better” is wrong, especially in heart/kidney disease.

EXAM TRAP**Chromium: retain the historical answer with a qualification**

F3 and O16 expect chromium as an enhancer of insulin action, matching the 2016 book. Current NIH ODS discusses uncertainty about chromium's essentiality and physiological role. The old association is not an endorsement of chromium supplements for diabetes.

Adult zinc RDA is 11 mg/day for men and 8 mg/day for women; 40 mg/day is the adult U.S. UL, not the recommended intake. F13's sex-unspecified “adults” wording is incomplete. A banana is not a strong zinc source compared with the other D11 options.

CHECK YOURSELF

- Someone takes large zinc doses and develops anemia. Why think beyond iron intake?**
Excess zinc can impair copper absorption; copper is needed for normal iron handling and other functions. Supplement interactions can create a second deficiency.

Past-paper distinction: element, carrier, store and function

NAME	WHAT IT ACTUALLY REPRESENTS	CONTRAST
Ceruloplasmin	A major copper-containing plasma protein, involved in iron metabolism	F19's 'stored by' wording is imprecise. Copper is the association, not evidence that plasma protein is the principal storage site.
Transferrin / ferritin	Iron transport / iron storage-related protein	These are not alternate names for ceruloplasmin. Inflammation complicates ferritin interpretation.
Phosphate	Structural component of nucleic acids, ATP and phospholipids	DNA/RNA backbone asks phosphorus, not the metal in hemoglobin.
Selenium / vitamin E	Selenoenzymes help remove peroxides; E interrupts lipid oxidative chains	Complementary roles are not proof that megadoses improve health.

Keshan disease is an endemic cardiomyopathy associated with very low selenium availability; infection and other factors can contribute. It is not a label for every cardiomyopathy. Zinc deficiency can link poor wound repair, altered taste and alopecia; prolonged excessive zinc can impair copper status. For food comparisons, match portion and preparation: edible-bone fish and fortified products make a blanket 'dairy always contains most calcium' claim unreliable.

CHECK YOURSELF

- 1 **A stem asks the mineral associated with ceruloplasmin, then asks the circulating transporter of iron. Give distinct answers.**

Copper is associated with ceruloplasmin; transferrin transports iron. Do not exchange the element, its associated protein and the storage concept.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 248-260, 266-271 (PDF counter, not printed folio) · F - nutrition final exam.pdf - 24 English screenshot pages - pp. 3, 6, 13-14, 19; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q11, Q19-20, Q23; question evidence, not an official answer key · NIH ODS - Zinc - Checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/Zinc-HealthProfessional/> · NIH ODS - Chromium - Checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/Chromium-Consumer/> · NIH ODS - Copper: Health Professional Fact Sheet - Introduction, copper status and zinc interactions; checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/Copper-HealthProfessional/>

18 / Pregnancy and lactation

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

Pregnancy increases several nutrient needs more than it increases energy needs, so nutrient density matters. Preconception status matters because early neural-tube development occurs before many pregnancies are recognized. A prenatal multivitamin is not a substitute for adequate food, and more is not always safer.

ISSUE	STUDY VALUE / PRINCIPLE	EXAM GUARDRAIL
Energy	Book convention: no routine increment in trimester 1; about +340 kcal/day in trimester 2; +450 in trimester 3	Increment above baseline, not the entire daily requirement. N25's arithmetic near +452 represents the same approximate third-trimester increment.
Protein / carbohydrate	Conventional pregnancy references: about +25 g protein/day above nonpregnant need; carbohydrate RDA 175 g/day	Not a reason to use high-dose protein supplements or severely restrict carbohydrate without supervision.
Iron / folate / B12	27 mg iron; 600 micrograms folate DFE; 2.6 micrograms B12 daily	Different units; folic-acid prevention guidance is not identical to total DFE.
Calcium / vitamin D	Adult pregnancy calcium 1,000 mg/day; adolescent pregnancy 1,300 mg/day; D 600 IU/day	Calcium recommendation follows age; pregnancy does not simply double every RDA.
Normal-BMI singleton weight gain	About 11.5-16 kg (25-35 lb)	Depends on prepregnancy BMI and singleton versus multiple gestation. N10's 11.3-15.8 kg is an approximate pound conversion.
Fish	FDA/EPA: 2-3 adult servings/week from lower-mercury Best Choices, totaling 8-12 oz	Species, mercury category and serving size matter; not a command to avoid all fish.

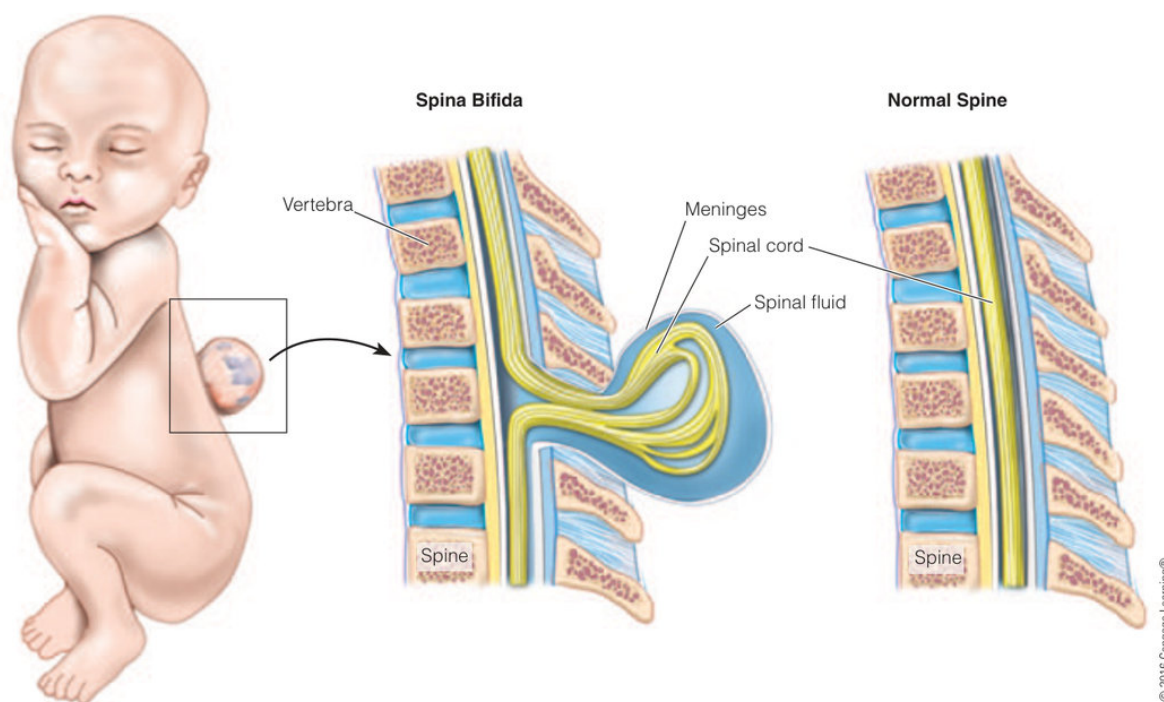


FIGURE 11-5 Spina bifida—A Neural Tube Defect

Spina bifida, a common neural tube defect, occurs when the vertebrae of the spine fail to close around the spinal cord, leaving it unprotected. The B vitamin folate—consumed prior to and during pregnancy—helps prevent spina bifida and other neural tube defects.

Figure 7 Folate-related prevention must begin early. This illustration shows one spinal neural-tube-defect pattern; not every neural tube defect looks like this.

Source: DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016), PDF p. 316

- Avoid excess preformed vitamin A; large liver/supplement exposures can be hazardous. Normal vegetable carotenoids should not be equated with retinoid megadoses.
- Alcohol exposure during pregnancy can harm development. Food safety and individual medical conditions also matter; do not derive treatment decisions from a multiple-choice screenshot.
- Lactation has its own energy and micronutrient requirements; it is not simply a continuation of every pregnancy number. Milk production and maternal stores must be considered.

CHECK THE SOURCE

Breastfeeding contraindications are context-specific

Classic infant galactosemia prevents ordinary breast-milk feeding. Untreated brucellosis can require interruption. With untreated active tuberculosis, direct contact and expressed milk are separate questions. “Extrapulmonary TB” or “lithium” alone does not provide enough detail for a universal yes/no rule. D28 needs the lecturer’s context and contemporary clinical guidance, not an unchecked historic key.

CHECK YOURSELF**1 Why is starting folic acid only late in pregnancy not equivalent to periconceptional prevention?**

Neural-tube closure occurs very early. Later adequacy still matters for maternal/fetal health but cannot retrospectively prevent an already established closure defect.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 311-319, 327-335 (PDF counter, not printed folio) · F - nutrition final exam.pdf - 24 English screenshot pages - pp. 1, 18; question evidence, not an official answer key · N - Nutrition .pdf - 28 general-nutrition scan pages - pp. 10, 13, 22-23, 25; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q25, Q28; question evidence, not an official answer key · NIH ODS - Iron - Checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/Iron-HealthProfessional/> · NIH ODS - Folate - Checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/Folate-HealthProfessional/> · FDA - Advice about eating fish - Checked 14 September 2026 - <https://www.fda.gov/food/consumers/advice-about-eating-fish> · NICHD - Healthy pregnancy - Checked 14 September 2026 - <https://www.nichd.nih.gov/health/topics/preconceptioncare/conditioninfo/healthy-pregnancy> · CDC - Breastfeeding contraindications - Checked 14 September 2026 - <https://www.cdc.gov/breastfeeding-special-circumstances/hcp/contraindications/index.html>

19 / Infant and child nutrition

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

Infants have high nutrient needs relative to body size. Milk feeding transitions to complementary foods as development and nutrient needs change. Readiness includes appropriate head/trunk control and safe handling of textures; feeding is not determined by a calendar alone.

- Complementary foods usually begin around 6 months; breast milk or suitable infant formula remains important. Include iron-rich foods and a progressively varied, developmentally safe diet.
- Do not replace breast milk/formula with ordinary cow's milk as the main drink before 12 months. Nutrient composition, iron displacement and other risks differ.
- Current CDC advice avoids honey and fruit/vegetable juice before 12 months. This corrects the older book table that introduces diluted juice earlier.
- Whole nuts and hard/round foods are choking hazards. A traditional porridge schedule in a past paper is not permission to feed whole hazelnuts to an infant.
- Do not rank fruit allergy risk with certainty from the four D29 options alone. Age, preparation, actual allergy history and current introduction guidance matter.
- During childhood and adolescence evaluate growth trajectories, dietary pattern and development, not isolated adult BMI cutoffs. Iron, calcium and energy needs vary with growth and life stage.

CHECK THE SOURCE**Local food sequence is still a source gap**

D26 asks which of almond porridge, legumes, wheat sprouts or hazelnut starts earlier. The likely intended traditional sequence is almond porridge, but no matching lecture schedule was supplied. Keep this as a course-convention item, not universal feeding guidance.

CHECK YOURSELF**1 Why is a young infant's severe B12 deficiency possible when the mother has few symptoms?**

Low maternal B12 stores/intake can result in inadequate infant supply, and the infant has limited reserves and rapid neurological development. Maternal appearance does not establish infant adequacy.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 342-353, 362-375 (PDF counter, not printed folio) · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q23, Q26, Q28-29; question evidence, not an official answer key · CDC - Foods and drinks to avoid or limit - Checked 14 September 2026 - <https://www.cdc.gov/infant-toddler-nutrition/foods-and-drinks/foods-and-drinks-to-avoid-or-limit.html>

20 / Ageing: muscle, hydration, B12 and D

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

Ageing often reduces energy expenditure through lower activity and loss of lean mass. Micronutrient and protein adequacy do not fall in the same proportion, so simply shrinking every food portion can worsen nutritional quality. Assess chewing, swallowing, appetite, medications, isolation, finances and functional ability.

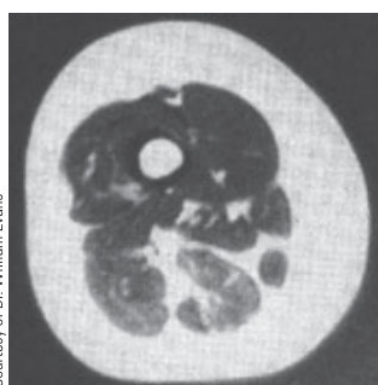
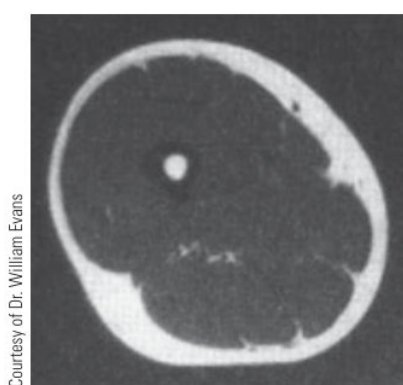


FIGURE 13-2 Sarcopenia

These cross sections of two women's thighs may appear to be about the same size from the outside, but the 20-year-old woman's thigh (left) is dense with muscle tissue. The 64-year-old woman's thigh (right) has lost muscle and gained fat, changes that may be largely preventable with strength-building physical activities.

Figure 8 Two cross-sections with different muscle/fat composition. Similar external size can conceal different tissue reserves.

Source: DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016), PDF p. 395

PROBLEM	WHY IT MATTERS	STUDY RESPONSE
Low lean mass / strength	Lower expenditure and functional reserve	Adequate protein/energy and suitable resistance activity; individual needs depend on health and renal status.
Reduced thirst / access	Dehydration may be overlooked	Evaluate fluid intake and losses; prescriptions differ if fluid restriction is indicated.
Food-bound B12 malabsorption	Reduced gastric acid and other disorders can reduce availability	A reliable absorbable source and assessment may be needed; eating meat does not rule out malabsorption.
Vitamin D risk	Less skin synthesis/exposure and altered metabolism	Reference need rises after 70; consider overall bone/muscle context.

PROBLEM	WHY IT MATTERS	STUDY RESPONSE
Constipation / oral problems	Reduced intake quality and comfort	Suitable fiber, fluids, mobility and clinical assessment; not laxative or supplement advice from a single stem.

CHECK THE SOURCE**Supplements are not automatic dementia or hearing-loss prevention**

D24, D27 and D30 ask about micronutrients and cognitive/auditory outcomes. Treat their selected answers as claims requiring the lecturer's supporting study. Correct a documented deficiency when appropriate, but do not turn an association into a general supplement prescription or a guaranteed preventive effect.

CHECK YOURSELF**1 Why may an older adult need more nutrient-dense choices even if daily energy needs decrease?**

A lower energy allowance leaves less room for nutrient-poor foods; needs for important micronutrients and maintenance of functional tissue remain.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 390-398 (PDF counter, not printed folio) · N - Nutrition .pdf - 28 general-nutrition scan pages - pp. 2, 7-8; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q24, Q27, Q30; question evidence, not an official answer key · NIH ODS - VitaminB12 - Checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/VitaminB12-HealthProfessional/>

21 / CVD, diabetes and renal diet principles

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

CONTEXT	PRIORITY PRINCIPLE	REJECT THIS OVERGENERALIZATION
DASH / hypertension	Vegetables, fruit, whole grains, suitable low-fat dairy, legumes/nuts and appropriate proteins; reduce sodium and excess saturated fat	It is not merely "low fat," nor useful only after hypertension is diagnosed.
Atherosclerotic risk	Assess overall risk and lipid pattern; replace saturated fats appropriately, include suitable fiber and support healthy activity/weight	Total cholesterol <200 mg/dL is not a universal complete treatment goal.
Diabetes	Individualized food pattern, carbohydrate amount/quality/distribution, activity, weight goals and medication context	Do not prescribe unlimited nuts/protein or universal weight loss regardless of nutritional status.
CKD without dialysis	Avoid excessive protein; an appropriate supervised plan may moderate protein while protecting nutritional status	"High creatinine" alone does not specify a complete diet or justify starving the patient.
Dialysis	Protein losses may increase protein needs; fluid/sodium/potassium/phosphorus advice depends on residual function, treatment and results	Dialysis does not make fluid or electrolyte intake unrestricted.

Nutrition assessment precedes advice. For the middle-aged person with diabetes in N14, evaluating and modifying the existing eating pattern and lifestyle is the best broad starting response among the options. Weight loss may help some patients, but it is not automatically appropriate for every person with diabetes.

EXAM TRAP**Marked kidney-diet answers can be unsafe**

N19 selects extra fluid to clear waste. That is not generally correct in kidney disease: fluid needs depend on urine output and volume status. N21 appropriately raises concern about fluid accumulation, but restrictions are individualized rather than universal.

Older CVD teaching may use saturated fat <7% of energy as a therapeutic target. Label this as the historical course/textbook convention. Current patient advice must consider the full clinical risk profile and current guidance, not only one legacy number. DASH commonly uses 2,300 mg sodium/day, with 1,500 mg lowering blood pressure further in the cited NHLBI guidance.

CHECK YOURSELF**1 Why might a protein restriction suitable before dialysis become inappropriate after dialysis starts?**

Dialysis removes amino acids/protein and changes nutritional demands. A plan must account for treatment losses and protein-energy wasting risk rather than carry an old restriction forward unchanged.

Past-paper mechanism: atherosclerosis, plaque and thrombosis

Atherosclerosis is plaque accumulation in arterial walls, not simply elevated cholesterol in a blood sample. Retained lipid and inflammatory cells contribute to foam cells and fatty streaks; a mature lesion contains lipid, cells and connective tissue and may calcify. Arteriosclerosis is the broader category of arterial thickening/hardening; atherosclerosis is one type.

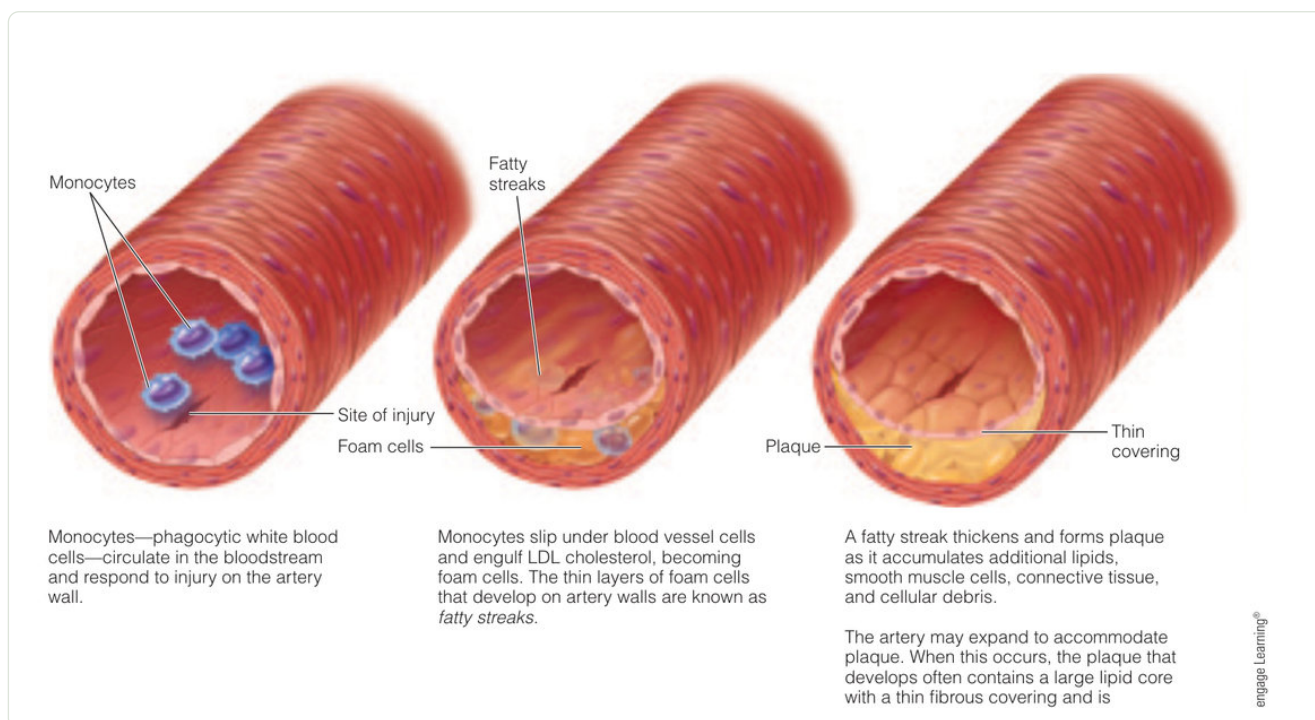


Figure 9 Read left to right: lipid-laden inflammatory cells, fatty streak, then plaque. The source illustrates progression; an individual lesion's behavior is not predicted solely by its size.

Identify Monocytes; Foam cells; Fatty streaks; Plaque; Thin covering

Source: DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016), PDF p. 602 (counter)

EVENT	WHAT CHANGES	EXAM CONSEQUENCE
Gradual narrowing	Reduced flow reserve through the affected artery	Supply may become inadequate when demand rises.
Plaque rupture with thrombosis	Thrombogenic material becomes exposed and a clot forms	Acute occlusion can occur without years of severe fixed narrowing.
Ischemia / infarction	Insufficient blood supply / tissue death from sustained injury	Do not call every transient reduction in flow an infarction.

CHECK YOURSELF

1 Why is 'raise HDL' or 'total cholesterol below one cutoff' not a complete response to an atherosclerosis question?

Atherosclerotic risk depends on the overall lipoprotein burden, smoking, blood pressure, diabetes and other factors. An isolated laboratory target does not describe plaque biology or constitute a complete treatment plan.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 576-585, 596-616, 630-643 (PDF counter, not printed folio) · N - Nutrition .pdf - 28 general-nutrition scan pages - pp. 9, 12, 14, 19, 21; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Q38; question evidence, not an official answer key · NHLBI - DASH - Checked 14 September 2026 - <https://www.nhlbi.nih.gov/health/dash-eating-plan> · NIDDK - Healthy eating in CKD - Checked 14 September 2026 - <https://www.nidk.nih.gov/health-information/kidney-disease/chronic-kidney-disease-ckd/healthy-eating-adults-chronic-kidney-disease> · Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 602-603, Fig. 22-3; PDF counter · NHLBI - What Is Atherosclerosis? - Disease definition and distinction from arteriosclerosis; checked 14 September 2026 - <https://www.nhlbi.nih.gov/health/atherosclerosis>

22 / Supplement: nutrition and oral health

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

CHECK THE SOURCE

Separate paper, unconfirmed course match

O is explicitly a Nutrition in Oral Health exam. The material below is supplementary and should not displace the general-nutrition core unless your lecturer confirms it belongs to your course. The final map accounts for all 36 questions, including those whose wording or key cannot be verified.

Dental caries is a biofilm-mediated process involving acid production from fermentable carbohydrates. Repeated exposure, retention on teeth, saliva and oral hygiene change risk. Caries is not identical to direct acid erosion: acidic drinks can erode enamel even without bacterial fermentation.

TERM / FACTOR	MECHANISM	EXAMPLES AND LIMITS
Cariogenic	Fermentable carbohydrate can support acid production and mineral loss	Sticky dried fruit, sweet drinks and frequent starchy/sugary snacks. Frequency and retention matter.
Cariostatic / noncariogenic	Does not substantially promote the cariogenic fermentation process under the comparison	Fish, many nonstarchy vegetables and nonfermentable sweeteners; preparations with added sugar differ.
Anticariogenic	Can actively help reduce cariogenic challenge	Cheese may stimulate saliva and provide calcium/phosphate/casein-related benefits. This is not a substitute for hygiene.
Saliva	Clearance, buffering, antimicrobial effects and mineral availability	Xerostomia removes several defenses. Sjögren syndrome and poorly controlled diabetes are relevant contexts.
Fluoride	Supports remineralization and reduces enamel susceptibility in suitable exposure	Excess during tooth formation can cause fluorosis.
Nutrient adequacy	Protein/energy, calcium/phosphorus, vitamin D and connective-tissue nutrients support oral tissues	No single deficiency sign proves a diagnosis, and periodontal disease is not cured by vitamin pills alone.

Vitamin C deficiency compromises connective tissue and can alter gums/dentin-related tissues. B-vitamin and iron deficiencies can produce oral mucosal findings. Such findings overlap with infection, trauma and other disorders, so the exam's single nutrient choice should not be used as a clinical diagnostic shortcut.

CHECK YOURSELF

- Why is sipping a sweet drink repeatedly different from consuming the same amount with a meal?**
Repeated exposure sustains more episodes of plaque acidification and may limit recovery; meal-associated saliva and clearance can reduce the duration of the challenge.

Oral-paper detail: know the tissue before choosing the nutrient

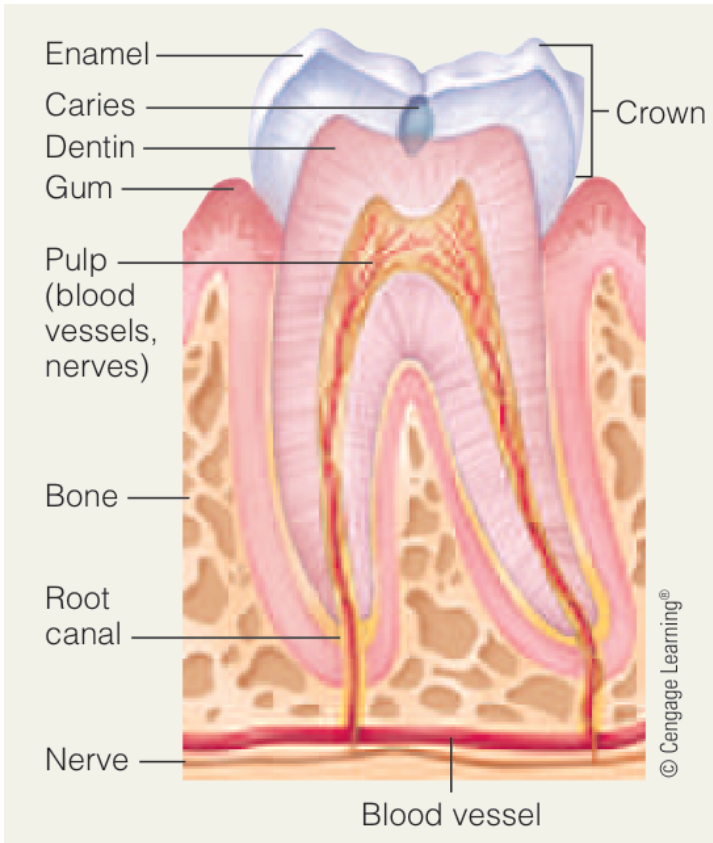


Figure 10 Original labeled tooth section. Enamel covers the crown; dentin surrounds the pulp. The pulp contains nerves and vessels; damage can progress beyond the enamel into deeper tissue.

Identify Enamel; Dentin; Pulp; Gum; Root canal; Bone; Nerve; Blood vessel

Source: DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016), PDF p. 512 (counter)

NUTRIENT / TISSUE	TEACHING FROM THE BOOK	BOUNDARY FOR THE EXAM
Protein-energy adequacy	Supports tooth development and salivary-gland structure/function	Deficiency can delay development and reduce defenses; O6 is not solely a calorie-calculation question.
Vitamin A	Supports epithelial differentiation, enamel development and salivary tissues	This does not validate O33's absolute 'first symptom' chronology.
Vitamin D, calcium and phosphorus	Support formation and mineralization of dental tissues	Low mineralization and low tissue mass are different concepts.
Vitamin C	Supports collagen-dependent connective tissues, including gingival/dentin-related support	O18/O29's exact odontoblast/pulp histology is not established by this general nutrition table.
Folate, zinc and other nutrients	Contribute to cell turnover, healing and host defenses	O14's selected combination is not an exclusive periodontal-treatment formula.

Dental plaque is a microbial biofilm; calculus is mineralized plaque. Gingivitis is inflammation of the gingiva, whereas periodontitis damages tooth-supporting tissues and can cause attachment and bone loss. A bleeding or fissured mouth lesion has nutritional and non-nutritional differentials. Riboflavin, iron and other deficiencies can overlap with local infection, irritation and dental disease. Do not diagnose a nutrient solely from a photograph or from O23's uncertain exception.

CHECK THE SOURCE**Do not turn incomplete dental keys into anatomy facts**

The precise claims in O4, O18, O23, O25, O29, O33 and O34 still require the original dental lecture or a defensible marking reference. The underlying tissues and nutritional mechanisms are now taught here, but this does not make 'B3 is unnecessary', 'magnesium never matters' or a fixed first-symptom rule true.

Oral-paper mechanism: acid challenge versus mineral recovery

COMPONENT	MECHANISM TO EXPLAIN	QUESTION LINK
Fermentable carbohydrate plus biofilm	Microbial acid production shifts balance toward mineral loss. Streptococcus mutans is a classic organism, but disease is an ecological biofilm process.	O9/O30: organism versus substrate.
Saliva	Clears substrates, buffers acid through bicarbonate/phosphate, provides mineral ions and antimicrobial components	O7/O24: less saliva means loss of several defenses, not merely less moisture.
Cheese / milk constituents	Calcium, phosphate and casein-related effects plus salivary stimulation/buffering	O19/O35: explain the mechanism; do not declare a food a cure.
Recaldent / CPP-ACP	Casein phosphopeptide-amorphous calcium phosphate; a milk-derived system holding calcium/phosphate in an available form	O2 tests identity. It is not a vitamin or a fluoride synonym. Milk-protein allergy is a product safety concern.
Saccharin / xylitol	Saccharin is noncariogenic; xylitol is not used like sucrose by cariogenic bacteria. Sugar-free gum can independently increase saliva.	O21/O32: separate replacement of fermentable sugar from gum-related saliva effects.
Fluoride	Supports mineral recovery and reduces susceptibility to acid challenge at appropriate exposure	Its mechanism differs from simply supplying energy or protein.

Caries requires the biofilm-related acid process; erosion can result from direct chemical acid exposure. Thus 'sugar-free' does not guarantee 'nonerosive'. Dried fruit, sticky starch and frequent sipping can prolong substrate contact; fish, broccoli or cheese may compare favorably only under the stated preparation and exposure. A universal ranking of grains, fruit or all foods is not justified by one paper's alternatives.

CHECK YOURSELF**1 A drink has no fermentable sugar but is strongly acidic. Is it harmless to enamel?**

Not necessarily. It may cause erosive tooth wear through direct acid action even if it provides less substrate for bacterial caries. Exposure frequency, saliva and the full product composition matter.

2 Expand CPP-ACP, then explain why it is not interchangeable with xylitol.

Casein phosphopeptide-amorphous calcium phosphate concerns mineral availability at the tooth surface. Xylitol substitutes for fermentable sugar, and a gum vehicle can stimulate saliva. These are different mechanisms; neither guarantees prevention or replaces dental care.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 512-516, 269-270 (PDF counter, not printed folio) · O - Taghziyeh.pdf - Nutrition in Oral Health paper - All 36 questions; question evidence, not an official answer key · Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 512-514; Fig. NP17-1 and Table NP17-1; PDF counter · American Dental Association - Nutrition and Oral Health - Caries, exposure frequency and erosion; checked 14 September 2026 - <https://www.ada.org/resources/ada-library/oral-health-topics/nutrition-and-oral-health> · GC - Tooth Mousse / Recaldent - Manufacturer identity and milk-protein allergy warning only, not independent efficacy ranking; checked 14 September 2026 - <https://www.gc.dental/europe/en-GF/products/toothmousse>

23 / Numerical checkpoint sheet

Source basis Past-paper topics cross-checked with the local textbook; explanatory book extensions are included

Unless stated otherwise, these are U.S./Canadian DRI teaching references for healthy adults, not deficiency-treatment doses. Age, sex, pregnancy/lactation and disease modify requirements. The exact framework requested by your lecturer takes priority for an examination question.

QUANTITY	VALUE	COMMON WRONG TURN
Carbohydrate / protein / fat / alcohol	4 / 4 / 9 / 7 kcal per gram	Swapping fat and alcohol or treating vitamins as energy.
Adult AMDR: carbohydrate / fat / protein	45-65% / 20-35% / 10-35% of total kcal	Using grams or a single food as the denominator.
Adult protein RDA	0.8 g/kg/day	Not 0.8 g/day; not every clinical target.
TEF, mixed diet	About 10% of TEE	Not a fixed fraction for every diet/person.
BMI	kg / m ²	Height in cm must be converted first.
B12 adult / pregnancy / lactation	2.4 / 2.6 / 2.8 micrograms per day	2 mg is not 2 micrograms.
Folate adult / pregnancy / lactation	400 / 600 / 500 micrograms DFE per day	DFE differs from folic-acid mass.
Iron man / woman 19-50 / pregnancy	8 / 18 / 27 mg/day	The sex-unspecified adult question is incomplete.
Zinc man / woman / adult UL	11 / 8 / 40 mg/day	UL is not RDA.
Vitamin D through 70 / over 70	600 / 800 IU per day	1 microgram = 40 IU.

QUANTITY	VALUE	COMMON WRONG TURN
Calcium 19-50 / adolescence	1,000 / 1,300 mg/day	Normal blood calcium does not prove adequate stores.
Vitamin A man / woman	900 / 700 micrograms RAE/day	Preformed retinol and carotenoids are not interchangeable by mass.
Vitamin C man / woman	90 / 75 mg/day; smokers +35 mg/day	Milligrams, not micrograms.
Pregnancy energy addition, trimesters 1/2/3	0 / ~340 / ~450 kcal/day in the book convention	Additional energy, not total daily intake.
Normal-BMI singleton gestational gain	~11.5-16 kg	Prepregnancy BMI and number of fetuses matter.

CHECK YOURSELF**1 At 2,400 kcal/day, what fat range corresponds to an adult AMDR of 20-35%?**

480-840 kcal from fat; divide by 9 = approximately 53-93 g/day. This calculates the range, not a personalized prescription.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 29-32, 119, 169-170, 216, 230-234, 256, 266, 315-319, 396 (PDF counter, not printed folio) · NIH ODS - VitaminB12 - Checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/VitaminB12-HealthProfessional/> · NIH ODS - Folate - Checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/Folate-HealthProfessional/> · NIH ODS - Iron - Checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/Iron-HealthProfessional/> · NIH ODS - Zinc - Checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/Zinc-HealthProfessional/> · NIH ODS - VitaminD - Checked 14 September 2026 - <https://ods.od.nih.gov/factsheets/VitaminD-HealthProfessional/>

24 / Integrated active recall: mechanism, not recognition

Source basis New synthesis exercises, not past-exam items

These are newly written review exercises based on the cited teaching, not recovered examination questions. Cover the answers, calculate with units, then explain why a tempting alternative is wrong.

CHECK YOURSELF**1 A meal contains 60 g carbohydrate, 20 g protein and 10 g fat. Find total energy and percent from fat.**

$240 + 80 + 90 = 410$ kcal; $90/410 \times 100 = 22.0\%$. Do not calculate 10/90 grams as an energy percentage.

2 A 60-kg healthy adult consumes 40 g protein/day. Compare with the conventional RDA.

$0.8 \times 60 = 48$ g/day; 40 g is below this planning reference. This alone does not prove clinical deficiency.

3 A diet provides 50% carbohydrate, 30% fat and 20% protein at 2,000 kcal. Convert to grams.

Carbohydrate $1,000/4 = 250$ g; fat $600/9 = 66.7$ g; protein $400/4 = 100$ g.

4 A chronic-inflammation patient has low serum iron and normal/high ferritin. Why not call this normal iron nutrition?

Inflammation can increase hepcidin and ferritin while restricting available iron. Concurrent absolute deficiency is still possible; interpret the full clinical/laboratory pattern.

5 An older person's BMI remains stable while strength falls. What does the scale miss?

Loss of muscle may be offset by gain in fat or fluid. Assess function, body composition, intake and illness.

6 Which two B-vitamin deficiencies can both raise homocysteine and cause megaloblastic anemia, but differ in methylmalonic acid?

Folate and B12; methylmalonic acid typically rises in B12 deficiency, not isolated folate deficiency. Kidney function also affects its concentration.

7 If renal activation of vitamin D falls, which conversion is impaired and why can the skeleton be affected?

25(OH)D to 1,25(OH)₂D through 1-alpha-hydroxylation. Reduced active-hormone effects can compromise mineral balance, within the broader disturbances of renal disease.

8 A PKU stem asks what to add and what to control. Give both, then one sweetener concern.

Tyrosine may need provision; phenylalanine must be carefully controlled in a specialist plan. Avoid aspartame because it supplies phenylalanine.

9 A prenatal product supplies 400 micrograms folic acid with food and the diet adds 200 micrograms natural folate. Approximate total DFE.

$400 \times 1.7 + 200 = 880$ micrograms DFE. This is a bioavailability calculation, not advice to add extra supplements.

10 Can normal serum calcium, normal BMI and a normal hemoglobin together exclude malnutrition?

No. Bone mineral can be depleted with normal circulating calcium; BMI can conceal low muscle; nutrient depletion can precede anemia. Match the measure to the question.

11 A protein-supplement advertisement promises muscle gain and a multivitamin promises more calories. Which assumptions fail?

Protein does not ensure hypertrophy without the broader training/energy context, and vitamins do not themselves provide metabolic calories.

12 A paper associates a vitamin deficiency with X-linked adrenoleukodystrophy. Does this establish its cause?

No. X-linked adrenoleukodystrophy is an ABCD1-related disorder of very-long-chain-fatty-acid handling. A marked vitamin association is not evidence of causation; ask for the lecturer's supporting source.

Past-paper reasoning: identify the missing condition

QUESTION FAMILY	CONDITION MISSING OR DISTORTED	WHAT REMAINS LEARNABLE
F4 / N1: numerical categories	Oligosaccharide cutoffs or a lecture-specific energy distribution	Know the stated convention; do not replace a range with a conveniently offered option.
F13 / F16	Sex for zinc RDA; the meaning of 'cyclic' for amino acids	11 versus 8 mg zinc; proline's ring to its amino nitrogen versus other aromatic rings.
N26 / D14	What is being measured; which drug is being used	Heat versus gas-exchange measurement versus prediction; specific nutrient-drug interactions instead of an invented universal ranking.
D21 / D23 / D24 / D27 / D30	Causation, study population or treatment evidence	A vitamin/mineral association does not prove genetic-disease causation or universal cognitive/hearing benefit.
D28 / D29	Infection/medicine context or allergy history and preparation	Clinical feeding decisions are not blanket rules inferred from old answer marks.
D36 / D40	Alpha- versus gamma-linolenic acid; absolute essential-fat intake	Alpha-linolenic is omega-3, gamma-linolenic omega-6; low total fat percentage alone is not proof of essential-fatty-acid deficiency.
O27	Regulator, date and drug formulation	A withdrawn drug, a combination product and its single ingredients are not interchangeable approval claims. Keep the original question ungraded.

CHECK YOURSELF

1 Why should an ambiguous source item stay visible even when it cannot be graded?

It reveals a tested topic and a possible lecturer convention, but a fabricated definitive key would teach false certainty. Learn the supported mechanism, identify the missing condition, and reserve the official answer until the original teaching or key resolves it.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 29-32, 111-120, 161-170, 218-234, 261-271, 315-319 (PDF counter, not printed folio) · MedlinePlus Genetics - X-linked adrenoleukodystrophy - Checked 14 September 2026 - <https://www.medlineplus.gov/genetics/condition/x-linked-adrenoleukodystrophy/>

25 / English final-exam screenshot map

Source basis Source-item accounting with editorial explanations, not official answer-key validation

Items are paraphrased for retrieval, not reproduced as an official key. SUPPORTED = the underlying concept is supported by the cited review sources; QUALIFIED = an important context/wording limit; CORRECTED = a marked value/claim needs correction; UNRESOLVED = do not memorize the marking without better evidence; DUPLICATE = repeated source stem.

ITEM / TOPIC	ANSWER AND REASONING	STATUS / REVIEW SECTION
F1 / Pregnancy iron	27 mg/day in the adult pregnancy DRI; distinguish dietary intake from a therapeutic dose.	SUPPORTED Read 16 / Iron: intake, transport, stores and anemia
F2 / Energy in fat	9 kcal/g; carbohydrate and protein supply 4 kcal/g.	SUPPORTED Read 07 / Energy calculations, BMR and fasting
F3 / Insulin-associated trace element	Chromium is the historical intended answer. Its essentiality and physiological role are uncertain in current NIH discussion; this is not a supplement recommendation.	QUALIFIED Read 17 / Water, major minerals and trace elements
F4 / Oligosaccharide length	Commonly 3-10 residues, but boundaries vary. The offered intervals do not give that conventional range exactly. Obtain the lecture definition rather than force an official key.	UNRESOLVED Read 03 / Carbohydrate, fiber and glycemic control
F5 / Not a monosaccharide	Lactose is a disaccharide: glucose plus galactose.	SUPPORTED Read 03 / Carbohydrate, fiber and glycemic control
F6 / Mineral cooperating with E	Selenium: peroxide-removing selenoenzymes complement vitamin E membrane antioxidant activity.	SUPPORTED Read 12 / Vitamins E and K: antioxidant versus clotting roles
F7 / Visual form of vitamin A	Retinal; distinguish retinoic-acid gene regulation.	SUPPORTED Read 10 / Vitamin A: vision, epithelium and toxicity
F8 / Thermic effect of food	Approximately 10% for a mixed diet. Composition and individual physiology modify the value.	QUALIFIED Read 07 / Energy calculations, BMR and fasting
F9 / Vitamin derived from cholesterol-related precursor	Vitamin D; skin 7-dehydrocholesterol is the immediate precursor for D3 synthesis.	SUPPORTED Read 11 / Vitamin D, calcium and mineralization
F10 / Antioxidant vitamin	E is the classic lipid-phase antioxidant among these choices. This does not mean other vitamins or carotenoids never participate in antioxidant processes.	SUPPORTED Read 12 / Vitamins E and K: antioxidant versus clotting roles

ITEM / TOPIC	ANSWER AND REASONING	STATUS / REVIEW SECTION
F11 / Adult B12 amount	Current adult DRI is 2.4 micrograms/day. The offered 2 micrograms is a rounded/older choice; 2 milligrams is not equivalent.	CORRECTED Read 14 / B6, folate and B12: connect the pathways
F12 / Thromboxane function	Platelet aggregation and vasoconstriction; the option "blood clotting" captures the intended association.	SUPPORTED Read 04 / Lipids, essential fatty acids and eicosanoids
F13 / Adult zinc amount	11 mg/day for men, 8 for women. 40 mg is the adult UL. The sex-unspecified stem is incomplete.	QUALIFIED Read 17 / Water, major minerals and trace elements
F14 / Mineral in DNA/RNA	Phosphorus in the phosphate backbone, also present in ATP and phospholipids.	SUPPORTED Read 17 / Water, major minerals and trace elements
F15 / Beriberi	Thiamin (B1) deficiency; neurological and cardiac manifestations differ by phenotype.	SUPPORTED Read 13 / B vitamins: enzymes and deficiency patterns
F16 / Cyclic amino acid	Proline is likely intended because its side chain closes onto its amino nitrogen. Phenylalanine and tryptophan also have rings, so "cyclic" alone is ambiguous.	QUALIFIED Read 05 / Protein, amino acids and nitrogen balance
F17 / Sweetest common sugar	Fructose among the listed sugars; perceived sweetness depends on test conditions.	SUPPORTED Read 03 / Carbohydrate, fiber and glycemic control
F18 / Neural tube defects	Folate inadequacy is the nutrient association; periconceptional folic acid reduces risk but does not explain every defect.	SUPPORTED Read 18 / Pregnancy and lactation
F19 / Ceruloplasmin-associated mineral	Copper. Ceruloplasmin is a major circulating copper-containing protein; the "stored by" wording is misleading.	CORRECTED Read 17 / Water, major minerals and trace elements
F20 / Phospholipid	Lecithin, commonly phosphatidylcholine. A fatty acid by itself is not a phospholipid.	SUPPORTED Read 04 / Lipids, essential fatty acids and eicosanoids
F21 / Renal vitamin D hydroxylation: ambiguous stem	Activation is renal 1-alpha-hydroxylation of 25(OH)D, but renal 24-hydroxylation also occurs. The original stem does not clearly restrict the question to activation, so a unique official key cannot be certified from organ location alone.	UNRESOLVED Read 11 / Vitamin D, calcium and mineralization
F22 / Smallest amino acid	Glycine; the side chain is hydrogen.	SUPPORTED Read 05 / Protein, amino acids and nitrogen balance
F23 / Pigeon-chest deformity	Vitamin D deficiency/rickets is the intended association. The shape alone is not specific for a nutritional diagnosis.	QUALIFIED Read 11 / Vitamin D, calcium and mineralization

ITEM / TOPIC	ANSWER AND REASONING	STATUS / REVIEW SECTION
F24 / Night blindness	Vitamin A deficiency is the classic nutritional cause.	SUPPORTED Read 10 / Vitamin A: vision, epithelium and toxicity

SOURCES F - nutrition final exam.pdf - 24 English screenshot pages - PDF pages 1-24; selected/marked answers are not an official key · DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - Exact topic-specific PDF locators and correction references are in the linked teaching sections 01-24

26 / Mixed-language scan map

Source basis Source-item accounting with editorial explanations, not official answer-key validation

Items are paraphrased for retrieval, not reproduced as an official key. SUPPORTED = the underlying concept is supported by the cited review sources; QUALIFIED = an important context/wording limit; CORRECTED = a marked value/claim needs correction; UNRESOLVED = do not memorize the marking without better evidence; DUPLICATE = repeated source stem.

ITEM / TOPIC	ANSWER AND REASONING	STATUS / REVIEW SECTION
N1 / Protein percentage of energy	10-15% is a conventional teaching distribution; the adult AMDR is 10-35%. No offered choice gives the full AMDR, so the intended lecture convention is unresolved.	UNRESOLVED Read 02 / Nutrients, DRIs and food choices
N2 / Older-adult protein assessment	Albumin is selected, but it is not a reliable standalone measure of intake or muscle stores. Use dietary history, trajectory, examination and illness context.	CORRECTED Read 08 / Assessment, BMI and body composition
N3 / Use of nonessential amino acids	Glucogenic amino-acid carbon can contribute to glucose synthesis. There is no dedicated inert amino-acid storage depot.	SUPPORTED Read 05 / Protein, amino acids and nitrogen balance
N4 / Anthropometry and CVD risk	Waist assesses central adiposity; BMI plus fat distribution adds context. No anthropometric measure alone is a complete CVD-risk assessment. Exact "best" option depends on the course comparison.	UNRESOLVED Read 08 / Assessment, BMI and body composition
N5 / Costochondral beading	Rachitic rosary in rickets; defective mineralization in a growing skeleton.	SUPPORTED Read 11 / Vitamin D, calcium and mineralization
N6 / Fiber and cholesterol excretion	Pectin, a viscous soluble fiber, among the given choices. Bile-acid losses help explain the cholesterol effect.	SUPPORTED Read 03 / Carbohydrate, fiber and glycemic control
N7 / Lower energy needs with ageing	Loss of lean/muscle tissue reduces resting expenditure; lower physical activity often contributes.	SUPPORTED Read 20 / Ageing: muscle, hydration, B12 and D
N8 / Nutrient concerns in older people	Vitamin D risk/need is important, particularly with reduced exposure/synthesis. Do not infer universally higher iron needs or lower fluid needs.	QUALIFIED Read 20 / Ageing: muscle, hydration, B12 and D

ITEM / TOPIC	ANSWER AND REASONING	STATUS / REVIEW SECTION
N9 / CVD dietary target	Saturated fat below 7% of energy is a historical therapeutic convention among the options. Overall risk, lipid management and the replacement nutrient matter; it is not a complete modern treatment target.	QUALIFIED Read 21 / CVD, diabetes and renal diet principles
N10 / Normal-BMI pregnancy weight gain	Approximately 11.5-16 kg for a singleton pregnancy. The selected 11.3-15.8 kg is an approximate conversion of 25-35 lb.	QUALIFIED Read 18 / Pregnancy and lactation
N11 / Central obesity at normal BMI	Abdominal fat distribution can confer elevated risk even when BMI is in the reference range. Individualize the response.	SUPPORTED Read 08 / Assessment, BMI and body composition
N12 / DASH pattern	A fruit/vegetable/whole-grain/appropriate-dairy pattern rich in fiber and key minerals. Relative protein/fat comparisons depend on the diet used for comparison.	QUALIFIED Read 21 / CVD, diabetes and renal diet principles
N13 / Fish during pregnancy	Current FDA/EPA guidance: 2-3 servings weekly from lower-mercury Best Choices, 8-12 oz total. A historical "twice" answer must not become an all-species rule.	QUALIFIED Read 18 / Pregnancy and lactation
N14 / First dietary response in diabetes	Assess and improve the existing diet/lifestyle. Do not automatically prescribe high protein, unlimited nuts or weight loss regardless of the patient.	SUPPORTED Read 21 / CVD, diabetes and renal diet principles
N15 / Iron deficiency outcome	Iron-deficiency anemia, not an inherited globin disorder such as sickle-cell disease or thalassemia.	SUPPORTED Read 16 / Iron: intake, transport, stores and anemia
N16 / Community health indicators	Indirect nutritional assessment: morbidity, mortality and social context do not directly measure an individual nutrient concentration.	SUPPORTED Read 08 / Assessment, BMI and body composition
N17 / Conditionally essential amino acid in PKU	Tyrosine, because phenylalanine-to-tyrosine conversion is impaired. Phenylalanine still requires controlled provision.	SUPPORTED Read 05 / Protein, amino acids and nitrogen balance
N18 / Fiber and cholesterol excretion	Duplicate of N6: pectin. Retained only for source traceability.	DUPLICATE Read 03 / Carbohydrate, fiber and glycemic control
N19 / Kidney disease diet	Extra fluid is not universally appropriate. Supervised protein moderation may fit nondialysis CKD, but creatinine alone does not specify protein/fluid needs.	CORRECTED Read 21 / CVD, diabetes and renal diet principles
N20 / Meaning of RDA	Intake sufficient for approximately 97-98% of healthy people in a defined group. EAR addresses 50%.	SUPPORTED Read 02 / Nutrients, DRIs and food choices
N21 / Dialysis and fluids	Avoid fluid accumulation; actual allowance depends on urine output, treatment and volume status. Dialysis may also increase protein needs.	QUALIFIED Read 21 / CVD, diabetes and renal diet principles

ITEM / TOPIC	ANSWER AND REASONING	STATUS / REVIEW SECTION
N22 / Physiological anemia	Pregnancy hemodilution: plasma volume expands more than red-cell mass. Athletes can also have dilutional patterns, so exact options/context matter.	QUALIFIED Read 16 / Iron: intake, transport, stores and anemia
N23 / Preconception multivitamin benefit	Neural-tube-defect risk reduction through adequate folic acid. Do not generalize to prevention of every neurological disorder.	SUPPORTED Read 18 / Pregnancy and lactation
N24 / Use of nonessential amino acids	Duplicate of N3: amino-acid carbon and gluconeogenesis.	DUPLICATE Read 05 / Protein, amino acids and nitrogen balance
N25 / Third-trimester energy increment	About +450 kcal/day above baseline (roughly +452 in the older calculation). This is an increment, not total daily energy.	QUALIFIED Read 18 / Pregnancy and lactation
N26 / Most accurate basal-energy assessment	A formula is a prediction. Direct calorimetry measures heat; indirect calorimetry uses respiratory gas exchange. Which measured method is intended depends on the wording.	CORRECTED Read 07 / Energy calculations, BMR and fasting
N27 / Exception among causes of iron-deficiency anemia	Chronic infection more typically causes anemia of inflammation. The marked chronic occult fecal bleeding is wrong: sustained blood loss is a classic iron-deficiency cause. Coexistence remains possible.	CORRECTED Read 16 / Iron: intake, transport, stores and anemia
N28 / Preventing iron-deficiency anemia	Suitable iron-rich foods plus vitamin C; separate tea from iron-rich meals where relevant. Do not give everyone iron supplements or prohibit all dairy.	SUPPORTED Read 16 / Iron: intake, transport, stores and anemia

SOURCES N - Nutrition .pdf - 28 general-nutrition scan pages - PDF pages 1-28; selected/marked answers are not an official key · DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - Exact topic-specific PDF locators and correction references are in the linked teaching sections 01-24

27 / Numbered bilingual compilation map

Source basis Source-item accounting with editorial explanations, not official answer-key validation

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ITEM / TOPIC	ANSWER AND REASONING	STATUS / REVIEW SECTION
D1 / Energy-yielding nutrients	Carbohydrate, fat and protein. Vitamins/minerals enable metabolism but do not yield calories.	SUPPORTED Read 02 / Nutrients, DRIs and food choices
D2 / Influence on food preference	Cultural heritage/family tradition is the expected category in this stem.	SUPPORTED Read 02 / Nutrients, DRIs and food choices

ITEM / TOPIC	ANSWER AND REASONING	STATUS / REVIEW SECTION
D3 / Whole grains, legumes, fruit and vegetables	These choices suggest a relatively low-fat, fiber-rich pattern among the options. Added fats and processing can change the composition.	QUALIFIED Read 02 / Nutrients, DRIs and food choices
D4 / Polysaccharides	Starch, glycogen and many fibers. Fiber also includes lignin, which is not a carbohydrate polymer.	QUALIFIED Read 03 / Carbohydrate, fiber and glycemic control
D5 / Fuel after severe depletion of other stores	Body protein is the intended answer. Protein already contributes earlier in fasting; the body does not wait for complete fat depletion.	QUALIFIED Read 05 / Protein, amino acids and nitrogen balance
D6 / Basal metabolism	Heartbeat and temperature maintenance, not walking or running.	SUPPORTED Read 07 / Energy calculations, BMR and fasting
D7 / Largest expenditure component	Basal/resting metabolism usually dominates in sedentary adults; highly active people can differ.	QUALIFIED Read 07 / Energy calculations, BMR and fasting
D8 / Index calculated from height and weight	BMI = kg/m ² . This does not measure fat distribution.	SUPPORTED Read 08 / Assessment, BMI and body composition
D9 / Incorrect dietary guideline	Replacing liquid unsaturated oils with solid saturated fats reverses the usual favorable substitution.	SUPPORTED Read 04 / Lipids, essential fatty acids and eicosanoids
D10 / Portion versus serving	A portion is an amount served/eaten; a serving is a defined reference amount. A portion is not necessarily standardized.	SUPPORTED Read 02 / Nutrients, DRIs and food choices
D11 / Poor zinc source among choices	Banana, compared with the stronger protein/mineral sources listed.	SUPPORTED Read 17 / Water, major minerals and trace elements
D12 / Vitamin promoting calcium/phosphate absorption	Vitamin D, through active hormonal effects.	SUPPORTED Read 11 / Vitamin D, calcium and mineralization
D13 / Vitamin linked to photophobia	Riboflavin (B2) is the classic association among the options; the symptom is nonspecific.	QUALIFIED Read 13 / B vitamins: enzymes and deficiency patterns
D14 / Vitamin most affected by drugs	Folate is marked. No universal ranking is established without naming the drug; learn folate/antifolate, B6/isoniazid and B12/metformin relationships.	UNRESOLVED Read 14 / B6, folate and B12: connect the pathways
D15 / Rich preformed vitamin A food	Liver. Compare meaningful serving sizes and consider pregnancy toxicity risk.	SUPPORTED Read 10 / Vitamin A: vision, epithelium and toxicity

ITEM / TOPIC	ANSWER AND REASONING	STATUS / REVIEW SECTION
D16 / Fatigue, anorexia, arrhythmia and distal tingling	Thiamin is the likely intended deficiency among the options. The cluster is not a diagnostic rule.	QUALIFIED Read 13 / B vitamins: enzymes and deficiency patterns
D17 / Vitamin E role	Antioxidant protection of lipids/membranes.	SUPPORTED Read 12 / Vitamins E and K: antioxidant versus clotting roles
D18 / Teratogenic excess	Preformed vitamin A/retinoids, not an instruction to avoid ordinary carotenoid-containing vegetables.	SUPPORTED Read 10 / Vitamin A: vision, epithelium and toxicity
D19 / Taste changes, alopecia and poor healing	Zinc deficiency is a classic association.	SUPPORTED Read 17 / Water, major minerals and trace elements
D20 / Lowest-potassium option	Cucumber is lower among these choices. Absolute food rankings require serving size and preparation.	QUALIFIED Read 17 / Water, major minerals and trace elements
D21 / Vitamin associated with adrenoleukodystrophy	Vitamin A is marked, but the association is not supported by the supplied explanatory material. X-linked disease is ABCD1-related, not established to be caused by a vitamin deficiency. Ask for the lecturer's original source.	UNRESOLVED Read 24 / Integrated active recall: mechanism, not recognition
D22 / Carnitine-synthesis vitamin	Vitamin C supports relevant hydroxylation reactions.	SUPPORTED Read 15 / Vitamin C, connective tissue and antioxidant logic
D23 / Mineral deficiency and attention symptoms	Zinc/iron is marked. Associations do not establish that all attention disorders are deficiencies or that supplements treat them; the intended study requires confirmation.	UNRESOLVED Read 17 / Water, major minerals and trace elements
D24 / Supplement and older-adult memory	Folate is marked. Routine supplementation is not a guaranteed cognitive preventive treatment; assess deficiency and exclude B12 problems. Course-specific study needed.	UNRESOLVED Read 20 / Ageing: muscle, hydration, B12 and D
D25 / Why excessive liver is a pregnancy concern	High preformed vitamin A exposure.	SUPPORTED Read 18 / Pregnancy and lactation
D26 / Earliest complementary food among choices	Almond porridge is marked, likely reflecting a local traditional sequence. No matching lecture schedule was supplied. Safe texture and age readiness are separate issues.	UNRESOLVED Read 19 / Infant and child nutrition
D27 / High-frequency hearing loss nutrient association	B12 is marked. A nutrient association does not establish a specific diagnosis or universal benefit from supplements. Confirm the lecturer's evidence.	UNRESOLVED Read 20 / Ageing: muscle, hydration, B12 and D

ITEM / TOPIC	ANSWER AND REASONING	STATUS / REVIEW SECTION
D28 / Exception to breastfeeding contraindications	Active extrapulmonary TB is marked. Disease site/infectivity, treatment and expressed-versus-direct feeding matter; lithium also needs individual specialist assessment. Do not treat the options as blanket rules.	QUALIFIED Read 18 / Pregnancy and lactation
D29 / Least allergenic fruit at 10 months	Pineapple is marked, but this universal ranking is unverified. Use the source only to identify the lecturer-specific question; do not infer an allergy guarantee.	UNRESOLVED Read 19 / Infant and child nutrition
D30 / Supplement preventing auditory-brainstem decline	B12 is marked. Prevention claim/study is unverified in the supplied course material. Correcting deficiency is not equivalent to universal supplementation.	UNRESOLVED Read 20 / Ageing: muscle, hydration, B12 and D
D31 / Healthy adult protein RDA	0.8 g/kg/day. Multiply by body mass; clinical requirements may differ.	SUPPORTED Read 05 / Protein, amino acids and nitrogen balance
D32 / Severe protein/antioxidant malnutrition	Kwashiorkor is intended. Modern explanation is multifactorial rather than isolated protein deficiency alone.	QUALIFIED Read 09 / Protein-energy malnutrition and refeeding
D33 / Amino acid precursor of niacin	Tryptophan.	SUPPORTED Read 13 / B vitamins: enzymes and deficiency patterns
D34 / Homocysteine in amino-acid metabolism	Methionine metabolism to cysteine via transsulfuration; B6 is relevant. Remethylation to methionine requires folate/B12.	SUPPORTED Read 14 / B6, folate and B12: connect the pathways
D35 / Fatty liver with immune impairment	Kwashiorkor is the intended malnutrition pattern; context distinguishes it from other fatty-liver diseases.	QUALIFIED Read 09 / Protein-energy malnutrition and refeeding
D36 / Not an omega-3 fatty acid	Oleic acid is omega-9. Alpha-linolenic acid is omega-3; "linolenic" without alpha/gamma is imprecise.	SUPPORTED Read 04 / Lipids, essential fatty acids and eicosanoids
D37 / Energy in 7 g fat plus 7 g protein	$63 + 28 = 91$ kcal.	SUPPORTED Read 07 / Energy calculations, BMR and fasting
D38 / Lipid-rich arterial plaque disease	Atherosclerosis.	SUPPORTED Read 21 / CVD, diabetes and renal diet principles
D39 / Fatty acid prominent in sunflower/safflower oils	Linoleic acid in conventional high-linoleic varieties. High-oleic cultivars are exceptions.	QUALIFIED Read 04 / Lipids, essential fatty acids and eicosanoids
D40 / Very-low-fat diet and essential-fat deficiency	Below 20% is marked and lies below the adult fat AMDR. Actual linoleic/alpha-linolenic intake determines adequacy; 20% is not a precise deficiency threshold.	QUALIFIED Read 04 / Lipids, essential fatty acids and eicosanoids

Continued from "27 / Numbered bilingual compilation map"

SOURCES D - Taghziyeh.docx - bilingual general-nutrition compilation - Questions 1-40; selected/marked answers are not an official key · DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - Exact topic-specific PDF locators and correction references are in the linked teaching sections 01-24

28 / Supplementary oral-health paper map

Source basis Source-item accounting with editorial explanations, not official answer-key validation

Items are paraphrased for retrieval, not reproduced as an official key. SUPPORTED = the underlying concept is supported by the cited review sources; QUALIFIED = an important context/wording limit; CORRECTED = a marked value/claim needs correction; UNRESOLVED = do not memorize the marking without better evidence; DUPLICATE = repeated source stem. This is a separate oral-health paper: relevance to your medical nutrition examination is unconfirmed.

ITEM / TOPIC	ANSWER AND REASONING	STATUS / REVIEW SECTION
01 / Exception among weight-gain disorders	Sjögren syndrome is the likely comparison with Cushing syndrome, hypothyroidism and adult growth-hormone deficiency. It is better known here for dry-mouth risk.	QUALIFIED Read 22 / Supplement: nutrition and oral health
02 / Remineralizing compound / product	Recaldent is CPP-ACP (casein phosphopeptide-amorphous calcium phosphate). Product identity is supported; this does not rank products or prescribe treatment.	SUPPORTED Read 22 / Supplement: nutrition and oral health
03 / Cariogenic food among choices	Dried mulberries: fermentable carbohydrate plus retention. Preparation and exposure frequency matter.	QUALIFIED Read 22 / Supplement: nutrition and oral health
04 / Nutrient not essential for tooth development	B3 is marked, alongside options C, A and iodine. The "not essential" wording is too absolute: normal development depends on general nutrient adequacy. No secure developmental exception is supplied.	UNRESOLVED Read 22 / Supplement: nutrition and oral health
05 / Iron in reproductive-age women versus men	18 and 8 mg/day respectively for the conventional healthy-adult groups. Pregnancy uses a different value.	SUPPORTED Read 16 / Iron: intake, transport, stores and anemia
06 / Oral effects of protein-energy malnutrition	Delayed eruption and salivary-gland dysfunction can occur; overall development and defense are impaired.	QUALIFIED Read 09 / Protein-energy malnutrition and refeeding
07 / Protective milk/cheese property	Buffering/reduction of oral acidity, alongside mineral and salivary effects.	SUPPORTED Read 22 / Supplement: nutrition and oral health
08 / Pregnancy supplement and neural-tube development	Folic acid; distinguish preventive folic-acid guidance from total folate DFE.	SUPPORTED Read 18 / Pregnancy and lactation
09 / Caries-associated organism	Streptococcus, classically <i>S. mutans</i> among the options. Caries is a biofilm/ecological process, not proof that one organism alone is sufficient.	QUALIFIED Read 22 / Supplement: nutrition and oral health

ITEM / TOPIC	ANSWER AND REASONING	STATUS / REVIEW SECTION
O10 / Age of infantile beriberi	2-5 months is marked, but the exact age window is a course convention rather than a universal boundary. Learn maternal/infant thiamin deficiency; verify the age key.	UNRESOLVED Read 13 / B vitamins: enzymes and deficiency patterns
O11 / Drug associated with weight gain	Prednisolone is the expected comparison. Glucocorticoids can increase appetite and alter metabolism; no drug should be changed from this study note.	QUALIFIED Read 08 / Assessment, BMI and body composition
O12 / Keshan disease mineral	Selenium deficiency association.	SUPPORTED Read 17 / Water, major minerals and trace elements
O13 / Less cariogenic fruit among choices	Watermelon is marked, plausibly reflecting water content/less retention than dried dates. No fruit ranking is independent of preparation, amount and exposure.	QUALIFIED Read 22 / Supplement: nutrition and oral health
O14 / Nutrients in periodontal health	Folate, zinc and vitamin C is the marked combination. These support cell turnover, healing and connective tissue; this is not a treatment recipe or an exclusive nutrient list.	QUALIFIED Read 22 / Supplement: nutrition and oral health
O15 / Wernicke-Korsakoff nutrient	Thiamin (B1).	SUPPORTED Read 13 / B vitamins: enzymes and deficiency patterns
O16 / Trace element and insulin sensitivity	Chromium is the historical answer; current essentiality/benefit is uncertain.	QUALIFIED Read 17 / Water, major minerals and trace elements
O17 / Components of DRI framework	EAR, RDA, AI and UL among the listed reference standards.	SUPPORTED Read 02 / Nutrients, DRIs and food choices
O18 / C deficiency during tooth development	Odontoblast degeneration is marked. Collagen/dentin support is a defensible mechanism, but the exact histological key needs the dental lecture reference.	UNRESOLVED Read 22 / Supplement: nutrition and oral health
O19 / Protective components in cheese	Calcium, phosphorus and casein-related components contribute to the explanation, with saliva/buffering.	SUPPORTED Read 22 / Supplement: nutrition and oral health
O20 / Cariostatic food	Fish among the options, assuming no added fermentable coating/sugar.	QUALIFIED Read 22 / Supplement: nutrition and oral health
O21 / Saccharin and caries	Noncariogenic sweetener; distinguish from fermentable sugar.	SUPPORTED Read 22 / Supplement: nutrition and oral health
O22 / Group at B12-deficiency risk	Older adults, particularly with impaired food-bound absorption. Other risk groups also exist.	SUPPORTED Read 20 / Ageing: muscle, hydration, B12 and D

ITEM / TOPIC	ANSWER AND REASONING	STATUS / REVIEW SECTION
O23 / Angular cheilitis nutrient exception	Zinc is marked, but oral lesions have overlapping nutritional/non-nutritional causes and the proposed exception is not secure. Ask for the original dental teaching table.	UNRESOLVED Read 22 / Supplement: nutrition and oral health
O24 / Xerostomia context	Poorly controlled diabetes can contribute to dry mouth and oral disease.	SUPPORTED Read 22 / Supplement: nutrition and oral health
O25 / Exception among enamel-solubility modifiers	Magnesium is marked. The precise contrast and mechanism are not adequately supported by the supplied explanatory sources; do not invent a universal enamel rule.	UNRESOLVED Read 22 / Supplement: nutrition and oral health
O26 / Cause of iron deficiency	Parasitic disease causing chronic blood loss, depending on organism and burden.	QUALIFIED Read 16 / Iron: intake, transport, stores and anemia
O27 / Weight-loss drug approval exception	Sibutramine is marked and was withdrawn in the U.S. for cardiovascular risk. The FDA-approved phentermine/topiramate combination is not the same as topiramate alone. Thus date/formulation make the single-answer wording ambiguous.	QUALIFIED Read 21 / CVD, diabetes and renal diet principles
O28 / Cariostatic food	Broccoli among the options; preparation matters.	QUALIFIED Read 22 / Supplement: nutrition and oral health
O29 / Dental development in C deficiency	Pulp changes are marked. General connective-tissue impairment is supported; the specific developmental histology needs the dental lecturer's source.	UNRESOLVED Read 22 / Supplement: nutrition and oral health
O30 / Cariogenic substrate	Fermentable carbohydrate, supporting bacterial acid production.	SUPPORTED Read 22 / Supplement: nutrition and oral health
O31 / Largest calcium source	Dairy is marked. Amounts depend on serving size and whether fish bones are eaten; compare matched portions instead of assigning an absolute food winner.	QUALIFIED Read 17 / Water, major minerals and trace elements
O32 / Anticariogenic option	Xylitol is marked. It is not fermented like sucrose by cariogenic bacteria; sugar-free gum can also stimulate saliva. This does not guarantee prevention or replace hygiene.	QUALIFIED Read 22 / Supplement: nutrition and oral health
O33 / Vitamin whose deficiency is not first noticed orally	A is marked. "First" presentation is variable and no oral sign alone diagnoses a deficiency. Learn the characteristic syndromes, not a rigid chronology.	UNRESOLVED Read 22 / Supplement: nutrition and oral health
O34 / Infant gum bleeding and maternal vitamin C	Excess maternal C is marked. Do not accept this as a diagnostic rule or established causal answer; neonatal bleeding requires proper assessment. Historical rebound-scurvy explanation needs evidence.	UNRESOLVED Read 15 / Vitamin C, connective tissue and antioxidant logic

ITEM / TOPIC	ANSWER AND REASONING	STATUS / REVIEW SECTION
O35 / Anticariogenic food among bread, cheese, apple, pasta	Cheese: saliva/buffering and mineral contributions.	SUPPORTED Read 22 / Supplement: nutrition and oral health
O36 / Most cariogenic food group	Grains is marked. Refined/sticky starches can be cariogenic, but frequency, retention, processing and added sugar prevent a universal ranking of entire food groups.	QUALIFIED Read 22 / Supplement: nutrition and oral health

SOURCES O - Taghziyeh.pdf - Nutrition in Oral Health paper - Questions 1-36; selected/marked answers are not an official key · DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - Exact topic-specific PDF locators and correction references are in the linked teaching sections 01-24 · NIDCR - Tooth decay process - Checked 14 September 2026; supports the specific product/mechanism distinction, not every marked answer - <https://www.nidcr.nih.gov/health-info/tooth-decay/more-info/tooth-decay-process> · NIDCR - Diabetes and oral health - Checked 14 September 2026; supports the specific product/mechanism distinction, not every marked answer - <https://www.nidcr.nih.gov/health-info/diabetes> · GC - Recaldent / Tooth Mousse product identity - Checked 14 September 2026; supports the specific product/mechanism distinction, not every marked answer - <https://www.gc.dental/ortho/en-XK/products/toothmousse> · FDA - Sibutramine withdrawal background - Checked 14 September 2026; supports the specific product/mechanism distinction, not every marked answer - <https://www.fda.gov/drugs/frequently-asked-questions-popular-topics/questions-and-answers-about-fdas-initiative-against-contaminated-weight-loss-products> · FDA - Phentermine/topiramate formulation - Checked 14 September 2026; supports the specific product/mechanism distinction, not every marked answer - https://www.accessdata.fda.gov/drugsatfda_docs/label/2024/022580s025lbl.pdf

29 / Final pass: secure marks and unresolved source requests

Source basis Editorial prioritization based on the available source questions

KEY IDEA

Your first revision pass
First master sections 02-05 (nutrient classification), 07 (calculations), 10-17 (vitamins/minerals) and 18-20 (life stages). Then do assessment/clinical diets (08 and 21). The oral supplement is lower priority until course relevance is confirmed.

DO WITHOUT LOOKING	THEN EXPLAIN WHY
4/4/9/7; grams-to-kcal; nutrient percentage; BMI	Why grams are not energy percentages and why BMI does not measure muscle.
Iron 8/18/27; B12 2.4; zinc 11/8; folate 400/600 DFE	Which population and units each number describes.
B1 beriberi; B2 oral/ocular; B3 pellagra; C scurvy; A night blindness; D rickets	The biochemical or tissue mechanism and one competing explanation.
PKU tyrosine; niacin tryptophan; homocysteine B6/B12/folate	Which conversion is impaired or requires the cofactor.
Liver 25-hydroxylation; kidney 1-alpha; dietary iron versus inflammatory anemia	Why a marked option can be wrong despite a familiar keyword.

DO WITHOUT LOOKING	THEN EXPLAIN WHY
Pregnancy, infant-feeding and older-adult principles	Why a context-specific guideline is not a universal supplement or food prohibition.

CHECK THE SOURCE**Bring these to the lecturer or obtain the missing slides**

Highest-priority unresolved general items: F4; N1 and N4; D14, D21, D23, D24, D26, D27, D29 and D30. Obtain definitions or the actual cited study/food schedule. The O map separately identifies unresolved dental items. No answer was invented to make the ledger look complete.

Accounting: 92 general-source items plus 36 oral-health-source items = 128 source items. N18 repeats N6; N24 repeats N3. Similar concepts recur across other files too, so these are not counts of unique exam concepts. The editable audit provides each item's topic, correction, source and linked section.

SOURCES N - Nutrition .pdf - 28 general-nutrition scan pages - See item-by-item maps F, N, D and O · F - nutrition final exam.pdf - 24 English screenshot pages - See item-by-item maps F, N, D and O · D - Taghziyeh.docx - bilingual general-nutrition compilation - See item-by-item maps F, N, D and O · O - Taghziyeh.pdf - Nutrition in Oral Health paper - See item-by-item maps F, N, D and O

30 / Past-paper gap review and completion checklist

Source basis Item-by-item source-ledger comparison against the teaching manuscript; author review, not an official key or independent clinical certification

All 128 original source records remain represented: F24, N28, D40 and O36. The existing maps in sections 25-28 give the individual topics and answer limitations. This pass adds the missing explanatory layer where an item had only a brief map answer. Scored-question eligibility in MED25 is a separate decision: a topic can be fully explained while the original wording remains ambiguous.

TEACHING UPDATE	ORIGINAL SOURCE ITEMS	WHAT CHANGED
11 / Vitamin D, calcium and mineralization	F9, F21, N5, D12	Activation was taught, but F21 was described too confidently as merely a misleading selected answer. Now: Past-paper distinction: activation is not every renal hydroxylation.
13 / B vitamins: enzymes and deficiency patterns	F15, D16, O10, O15	Beriberi and Wernicke features were present, but Korsakoff and the infant-maternal link were only named in the map. Now: Past-paper distinction: four thiamin-deficiency presentations.
17 / Water, major minerals and trace elements	F6, F14, F19, D19, O12, O16, O31	Mineral names were covered; carrier-versus-storage, Keshan context and comparison logic needed clearer teaching. Now: Past-paper distinction: element, carrier, store and function.
21 / CVD, diabetes and renal diet principles	D38, D9, N9, N11, N12	Risk-reduction principles were present, but D38's disease definition and plaque progression were only briefly mapped. Now: Past-paper mechanism: atherosclerosis, plaque and thrombosis.

TEACHING UPDATE	ORIGINAL SOURCE ITEMS	WHAT CHANGED
08 / Assessment, BMI and body composition	O1, O11, N2, N4, N11	The source map gave endocrine/drug answers without a side-by-side explanation in teaching notes. Now: Oral-paper crossover: weight change is not one mechanism.
22 / Supplement: nutrition and oral health	O4, O6, O14, O18, O19, O23, O25, O29, O33, O34	Oral nutrient roles were broad; the original book's tissue-development distinctions and labeled tooth diagram were absent. Now: Oral-paper detail: know the tissue before choosing the nutrient.
22 / Supplement: nutrition and oral health	O2, O7, O9, O13, O19, O20, O21, O24, O28, O30, O32, O35, O36, O3	The maps named Recaldent, sweeteners and organisms; teaching did not explain the product or compare these mechanisms in enough detail. Now: Oral-paper mechanism: acid challenge versus mineral recovery.
24 / Integrated active recall: mechanism, not recognition	F4, F13, F16, N1, N26, D14, D21, D23, D24, D27, D28, D29, D30, D36, D40, O27	The uncertain items were visible individually but there was no compact method for handling their missing conditions. Now: Past-paper reasoning: identify the missing condition.

CHECK THE SOURCE**What is not resolved by adding more notes**

No Nutrition lecture transcript or official current syllabus was supplied. Local food-introduction sequences, cognitive/hearing supplement claims, certain dental histology exceptions and date-dependent drug approvals still need their original lecture or official key. They remain visible in the maps and in MED25 Final Exam source review; none has been converted into an invented definitive answer.

Study order: general-nutrition teaching first, then the F/N/D source maps; use the oral-health supplement if your lecturer confirms it is in scope. Revisit missed topics by section title. The editable audit/nutrition-past-paper-teaching-audit.json records each source ID, teaching destination and added blocks; the previous reading copy is retained under audit/before-nutrition-past-paper-teaching-pass.

SOURCES DeBruyne, Pinna & Whitney - Nutrition & Diet Therapy, 9th ed. (2016) - PDF pp. 7-12, 23-32 (PDF counter, not printed folio) · F - nutrition final exam.pdf - 24 English screenshot pages - All 24 pages; question evidence, not an official answer key · N - Nutrition .pdf - 28 general-nutrition scan pages - All 28 pages; question evidence, not an official answer key · D - Taghziyeh.docx - bilingual general-nutrition compilation - Questions 1-40; question evidence, not an official answer key · O - Taghziyeh.pdf - Nutrition in Oral Health paper - Questions 1-36; question evidence, not an official answer key

APPENDIX A**Coverage limits and next sources**

- No official current nutrition syllabus, lecturer slide set or nutrition transcript was located in the inspected course folders. Confirm scope before treating book extensions as assigned.
- Past-paper interface selections and DOCX ticks are not verified official marking keys. Ambiguous, dated, incomplete or contradictory questions are flagged in the maps.
- The oral-health paper is a separate course context; its relevance to this medical nutrition exam is unconfirmed.
- Some exam items about infant food sequence, hearing/cognition supplements, oral developmental details and historical drug status need the lecturer's original source; no invented authoritative key is supplied.

- This pass involved source comparison, arithmetic checks, OCR/image checks and PDF layout inspection by the author; it is not an independent clinician review. Numerical recommendations are study references, not personal prescriptions.

APPENDIX B

Local source inventory

Exact paths and source hashes are stored in `_build/source-manifest.json`. Section source lines distinguish original evidence from book additions.

B - Nutrition - Nutrition & Diet Therapy, 9th ed. (extensionless PDF)
N - Nutrition .pdf - 28 general-nutrition scan pages
F - nutrition final exam.pdf - 24 English screenshot pages

D - Taghziyeh.docx - bilingual general-nutrition compilation (original Persian filename in source manifest)
O - Taghziyeh.pdf - Nutrition in Oral Health paper (original Persian filename in source manifest)